Differential Geometry 2

Problem 7: Orientations and change of variables on an orientable manifold

Let X be a connected orientable n-manifold.

(a) X has exactly two orientations; orientation sign of a diffeomorphism. Recall one convenient definition:

- An *orientation* on X is a choice, for every $x \in X$, of one of the two connected components of $\Lambda^n(T_x^*X) \setminus \{0\}$, varying continuously with x. Equivalently, it is a maximal atlas $\{(U_\alpha, \varphi_\alpha)\}$ whose transition maps have positive Jacobian determinant.

Because X is orientable, it admits at least one orientation; fix one and call it $\mathcal{O}$. Define its *opposite* orientation $-\mathcal{O}$ by reversing the choice at every point: a basis $(v_1, \dots, v_n)$ of T_xX is positively oriented for $-\mathcal{O}$ iff it is negatively oriented for $\mathcal{O}$.

We claim these are the only two orientations.

Claim. If $\mathcal{O}'$ is any other orientation on connected X, then $\mathcal{O}' = \mathcal{O}$ or $\mathcal{O}' = -\mathcal{O}$.
Proof. Fix a nowhere-zero n-form μ compatible with $\mathcal{O}$ on each coordinate patch (or globally if desired); locally it exists because orientations are represented by local volume forms, and we can work locally. On any connected coordinate chart $U \subset X$ choose an n-form μ_U that is positive for $\mathcal{O}$ on U. Similarly, choose ν_U that is positive for $\mathcal{O}'$ on U. Since both are nowhere zero on U, there exists a smooth function $f_U : U \to \mathbb{R} \setminus \{0\}$ with

$$\nu_U = f_U\, \mu_U.$$

Now f_U cannot change sign on U because U is connected and f_U never vanishes. Thus on U either ν_U has the same sign as μ_U everywhere (so $\mathcal{O}' = \mathcal{O}$ on U), or it has the opposite sign everywhere (so $\mathcal{O}' = -\mathcal{O}$ on U).

Consider the set

$$A := \{x \in X : \mathcal{O}'_x = \mathcal{O}_x\}.$$

In local oriented coordinates, the sign relation $\nu = f\mu$ shows A is open (sign locally constant) and its complement is also open. Since X is connected, A is either empty or all of X. Hence $\mathcal{O}' = \mathcal{O}$ everywhere or $\mathcal{O}' = -\mathcal{O}$ everywhere. □

Therefore a connected orientable manifold has exactly two orientations.

Orientation-preserving vs. reversing diffeomorphisms. Let $F : X \to X$ be a diffeomorphism. Fix an orientation $\mathcal{O}$ on X. At each point x, the derivative $dF_x : T_xX \to T_{F(x)}X$ maps oriented bases to bases. We say:

- F is *orientation-preserving* if dF_x sends positively oriented bases at x (for $\mathcal{O}$) to positively oriented bases at $F(x)$ (for $\mathcal{O}$), for all x.
- F is *orientation-reversing* if it sends positively oriented bases to negatively oriented bases, for all x.

Equivalently, in any oriented chart, the Jacobian determinant $\det(D(\varphi \circ F \circ \varphi^{-1}))$ is positive everywhere (preserving) or negative everywhere (reversing). Since $\det(dF_x) \neq 0$ and depends continuously on x, on connected X its sign is constant, so the dichotomy is global.

(b) Show $\int_X F^*\omega = \pm \int_X \omega$ for compactly supported n-forms. Let $\omega \in \Omega^n_c(X)$ be a compactly supported smooth n-form. Because F is a diffeomorphism, $F^*\omega$ is also compactly supported. Fix an oriented atlas $\{(U_\alpha, \varphi_\alpha)\}$ compatible with the chosen orientation $\mathcal{O}$ and a smooth partition of unity $\{\rho_\alpha\}$ subordinate to $\{U_\alpha\}$. Then

$$\int_X F^*\omega = \sum_\alpha \int_X \rho_\alpha\, F^*\omega = \sum_\alpha \int_{U_\alpha} \rho_\alpha\, F^*\omega.$$

Set $\eta_\alpha := \rho_\alpha\, \omega$, so η_α has compact support in U_α and $\omega = \sum_\alpha \eta_\alpha$. Hence it suffices to prove the statement when ω is supported in a single oriented chart.

So assume $\operatorname{supp}(\omega) \subset U$ where (U, φ) is an oriented chart with $\varphi : U \to V \subset \mathbb{R}^n$ a diffeomorphism. Write ω in these coordinates as

$$\omega|_U = f(x)\, dx^1 \wedge \cdots \wedge dx^n, \qquad x = (x^1, \ldots, x^n) \in V,$$

where $f \in C^\infty_c(V)$ and $dx^1 \wedge \cdots \wedge dx^n$ is the standard volume form on $\mathbb{R}^n$. Define the local representative of F:

$$\widetilde{F} := \varphi \circ F \circ \varphi^{-1} :\ V \to V' \subset \mathbb{R}^n.$$

Then (standard pullback computation)

$$F^*\omega = (f \circ \widetilde{F})\ \det(D\widetilde{F})\, dx^1 \wedge \cdots \wedge dx^n \quad \text{on } U.$$

Therefore

$$\int_U F^*\omega = \int_V (f \circ \widetilde{F}(x))\ \det(D\widetilde{F}(x))\, dx.$$

Now apply the Euclidean change-of-variables theorem for diffeomorphisms:

$$\int_V (f \circ \widetilde{F})\ \det(D\widetilde{F})\, dx = \int_{\widetilde{F}(V)} f(u)\, du \quad \text{if } \det(D\widetilde{F}) > 0,$$

and

$$\int_V (f \circ \widetilde{F})\ \det(D\widetilde{F})\, dx = -\int_{\widetilde{F}(V)} f(u)\, du \quad \text{if } \det(D\widetilde{F}) < 0,$$

because $\det(D\widetilde{F})$ has constant sign and the standard formula uses $|\det|$; equivalently, $\det = \operatorname{sign}(\det)\,|\det|$.

Since F is a diffeomorphism of the manifold, the sign of $\det(D\widetilde{F})$ is exactly the global orientation sign

$$\operatorname{sgn}(F) = \begin{cases} +1 & \text{if } F \text{ is orientation-preserving,} \\ -1 & \text{if } F \text{ is orientation-reversing.} \end{cases}$$

Also, $\widetilde{F}(V) = \varphi(F(U))$ and F is bijective, so integrating f over $\widetilde{F}(V)$ corresponds to integrating ω over $F(U)$, and because ω is supported in U we have $\int_{F(U)} \omega = \int_X \omega$. Thus on this chart we ob

The sphere S^2: tangent spaces and local coordinates (charts)

Sphere as a submanifold of $\mathbb{R}^3$. Consider the unit sphere

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : \; x^2 + y^2 + z^2 = 1\}.$$

It is a smooth 2-dimensional manifold embedded in $\mathbb{R}^3$.

Tangent space at a point. Let $p \in S^2$. A convenient characterization of the tangent space is

$$T_pS^2 = \{v \in \mathbb{R}^3 : \; v \cdot p = 0\}.$$

That is, tangent vectors are precisely those vectors orthogonal to the radius vector at p.

Example at the north pole. At $N = (0, 0, 1)$ we have

$$T_NS^2 = \{(a, b, c) \in \mathbb{R}^3 : \; c = 0\},$$

the xy-plane.

Why we need charts: no single global coordinate system on S^2

Local coordinates (charts). A *chart* on S^2 is a diffeomorphism from an open set $U \subset S^2$ to an open set in $\mathbb{R}^2$. One cannot cover all of S^2 by a single smooth chart; instead one uses a collection of charts whose domains cover S^2.

(A) Spherical coordinates (θ, ϕ)

Parametrization. Define

$$F(\theta, \phi) = (\sin\theta\cos\phi, \; \sin\theta\sin\phi, \; \cos\theta), \qquad 0 < \theta < \pi, \;\; 0 < \phi < 2\pi.$$

This parametrizes S^2 minus the north and south poles (where ϕ is not well-defined).

Tangent basis from the parametrization. At $p = F(\theta, \phi)$, the partial derivatives span the tangent plane:

$$F_\theta(\theta, \phi), \quad F_\phi(\theta, \phi) \;\in\; T_{F(\theta,\phi)}S^2,$$

and (away from the poles) they form a basis of $T_{F(\theta,\phi)}S^2$.

Forms in spherical coordinates. On this chart, any 1-form can be written as

$$\alpha = A(\theta,\phi)\,d\theta + B(\theta,\phi)\,d\phi,$$

and any 2-form as

$$\omega = f(\theta,\phi)\,d\theta \wedge d\phi.$$

The standard area form on the unit sphere is

$$\omega_{S^2} = \sin\theta\,d\theta \wedge d\phi, \qquad \int_{S^2} \omega_{S^2} = 4\pi.$$

(B) Stereographic coordinates (u, v) (charts covering S^2 by two maps)

Stereographic projection from the north pole. Let $N = (0,0,1)$. The stereographic projection from N maps

$$\pi_N : S^2 \setminus \{N\} \to \mathbb{R}^2, \qquad \pi_N(x,y,z) = \left(\frac{x}{1-z}, \frac{y}{1-z}\right).$$

This provides smooth coordinates (u, v) on S^2 minus the north pole.

Stereographic projection from the south pole. Similarly, from $S = (0,0,-1)$ one defines

$$\pi_S : S^2 \setminus \{S\} \to \mathbb{R}^2, \qquad \pi_S(x,y,z) = \left(\frac{x}{1+z}, \frac{y}{1+z}\right),$$

giving coordinates on S^2 minus the south pole. Together the two charts cover all of S^2.

Charts (TikZ sketches)

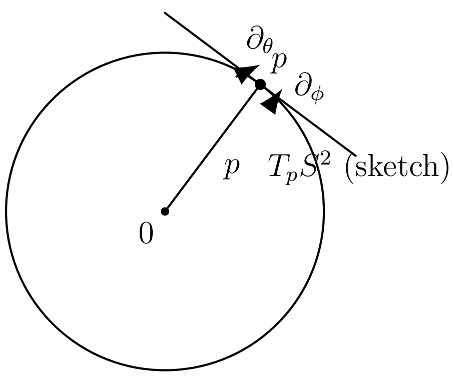


Figure 1: Conceptual picture: a point $p \in S^2$, the radius vector, and the tangent plane T_pS^2 (drawn as a tangent line in 2D).

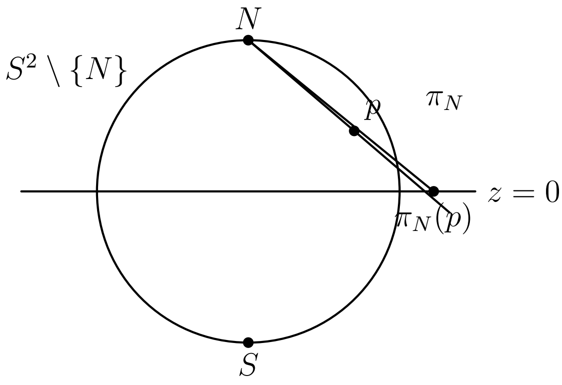


Figure 2: Stereographic projection from the north pole N: a chart $\pi_N : S^2 \setminus \{N\} \to \mathbb{R}^2$ (plane $z = 0$).

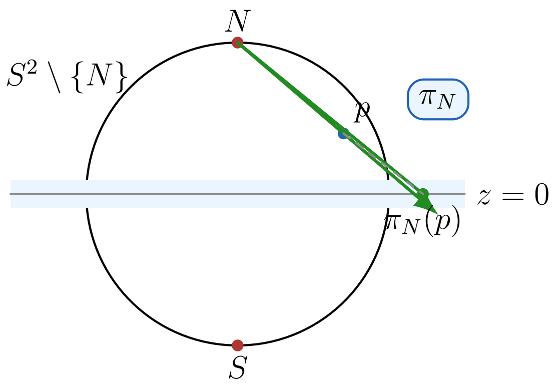


Figure 3: Stereographic projection from the north pole N: the chart $\pi_N : S^2 \setminus \{N\} \to \mathbb{R}^2$ (target plane $z = 0$).

Problem 8: Antipodal map, orientability of real projective space, and top de Rham cohomology

(a) For which n is the antipodal map $\alpha : S^n \to S^n$ orientation-preserving?

(b) By considering a natural $2 : 1$ map $\pi : S^n \to \mathbb{RP}^n$, deduce the values of n for which $\mathbb{RP}^n$ is orientable. You may assume without proof that π is locally a diffeomorphism.

(c) By combining these ideas with Q5 and Q7, compute $H^n_{\mathrm{dR}}(\mathbb{RP}^n)$ for all n.

Problem 8. Antipodal map, orientability of real projective space, and top de Rham cohomology

Solution. Let $\alpha : S^n \to S^n$ be the antipodal map $\alpha(x) = -x$.

(a) Consider the linear map $-I : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$. Its determinant is

$$\det(-I) = (-1)^{n+1}.$$

The antipodal map is the restriction of $-I$ to S^n, and its degree equals

$$\deg(\alpha) = (-1)^{n+1}.$$

A smooth map $S^n \to S^n$ is orientation-preserving iff its degree is $+1$. Hence α is orientation-preserving iff

$$(-1)^{n+1} = +1 \quad \Longleftrightarrow \quad n \text{ is odd.}$$

(b) Let $\pi : S^n \to \mathbb{RP}^n$ be the natural $2:1$ quotient map. We may assume π is locally a diffeomorphism, hence a covering map. The nontrivial deck transformation is precisely the antipodal map α.

An orientation on $\mathbb{RP}^n$ pulls back via π to an orientation on S^n; conversely, an orientation on S^n descends to $\mathbb{RP}^n$ iff it is invariant under deck transformations. Therefore

$$\mathbb{RP}^n \text{ is orientable} \iff \alpha \text{ is orientation-preserving on } S^n \iff n \text{ is odd.}$$

(c) With real coefficients, torsion vanishes in cohomology. Also $\mathbb{RP}^n$ is connected, so

$$H^0_{\mathrm{dR}}(\mathbb{RP}^n) \cong \mathbb{R}.$$

For $0 < k < n$, the (integral) cohomology of $\mathbb{RP}^n$ consists only of 2-torsion, hence

$$H^k_{\mathrm{dR}}(\mathbb{RP}^n) \cong 0 \qquad (0 < k < n).$$

Finally, the top de Rham cohomology satisfies

$$H^n_{\mathrm{dR}}(\mathbb{RP}^n) \cong \begin{cases} \mathbb{R}, & \mathbb{RP}^n \text{ orientable,} \\ 0, & \mathbb{RP}^n \text{ non-orientable,} \end{cases}$$

so using part (b),

$$H^n_{\mathrm{dR}}(\mathbb{RP}^n) \cong \begin{cases} \mathbb{R}, & n \text{ odd,} \\ 0, & n \text{ even.} \end{cases}$$

Equivalently, for all k,

$$H^k_{\mathrm{dR}}(\mathbb{RP}^n) \cong \begin{cases} \mathbb{R}, & k = 0, \\ 0, & 0 < k < n, \\ \mathbb{R}, & k = n \text{ and } n \text{ odd,} \\ 0, & k = n \text{ and } n \text{ even.} \end{cases}$$

Problem 8. Antipodal map, orientability of real projective space, and top de Rham cohomology

Solution. Let $\alpha : S^n \to S^n$ be the antipodal map $\alpha(x) = -x$.

(a) The map α is the restriction to $S^n \subset \mathbb{R}^{n+1}$ of the linear map $-I$ on $\mathbb{R}^{n+1}$. Its determinant is

$$\det(-I) = (-1)^{n+1}.$$

The antipodal map has degree

$$\deg(\alpha) = (-1)^{n+1}.$$

A smooth map $S^n \to S^n$ is orientation-preserving iff its degree is $+1$. Hence

$$\alpha \text{ is orientation-preserving} \iff (-1)^{n+1} = +1 \iff n \text{ is odd.}$$

(b) Let $\pi : S^n \to \mathbb{RP}^n$ be the natural $2:1$ quotient map. We may assume π is locally a diffeomorphism, hence a covering map. The only nontrivial deck transformation is the antipodal map α.

An orientation on $\mathbb{RP}^n$ pulls back along π to an orientation on S^n. Conversely, an orientation on S^n descends to $\mathbb{RP}^n$ iff it is invariant under all deck transformations, i.e. iff α preserves orientation. Therefore

$$\mathbb{RP}^n \text{ is orientable} \iff \alpha \text{ is orientation-preserving on } S^n \iff n \text{ is odd.}$$

(c) Since $\mathbb{RP}^n$ is connected, we have

$$H^0_{\mathrm{dR}}(\mathbb{RP}^n) \cong \mathbb{R}.$$

For $0 < k < n$, the (integral) cohomology of $\mathbb{RP}^n$ is 2-torsion only, hence with real coefficients (and therefore in de Rham cohomology) it vanishes:

$$H^k_{\mathrm{dR}}(\mathbb{RP}^n) = 0 \qquad (0 < k < n).$$

Finally, for a closed connected n-manifold M,

$$H^n_{\mathrm{dR}}(M) \cong \begin{cases} \mathbb{R}, & M \text{ orientable,} \\ 0, & M \text{ non-orientable.} \end{cases}$$

Using part (b), $\mathbb{RP}^n$ is orientable iff n is odd, hence

$$H^n_{\mathrm{dR}}(\mathbb{RP}^n) \cong \begin{cases} \mathbb{R}, & n \text{ odd,} \\ 0, & n \text{ even.} \end{cases}$$

Equivalently, for all k,

$$H^k_{\mathrm{dR}}(\mathbb{RP}^n) \cong \begin{cases} \mathbb{R}, & k = 0, \\ 0, & 0 < k < n, \\ \mathbb{R}, & k = n \text{ and } n \text{ odd,} \\ 0, & k = n \text{ and } n \text{ even.} \end{cases}$$

Problem 9: Green's theorem via Stokes' theorem (pullback of a 1-form)

Let Σ be a compact 2-manifold-with-boundary, let $F : \Sigma \to \mathbb{R}^2$ be a smooth map, and let α be the 1-form $P\,dx + Q\,dy$ on $\mathbb{R}^2$ (where P and Q are smooth functions). Prove a version of Green's theorem by applying Stokes to $F^*\alpha$.

Problem 9. Green's theorem via Stokes' theorem (pullback of a 1-form)
Solution. Let Σ be a compact oriented 2-manifold with boundary, let $F : \Sigma \to \mathbb{R}^2$ be smooth, and let

$$\alpha = P\,dx + Q\,dy$$

be a 1-form on $\mathbb{R}^2$, where P, Q are smooth functions.

Apply Stokes' theorem to the pulled-back 1-form $F^*\alpha$ on Σ:

$$\int_{\partial\Sigma} F^*\alpha = \int_{\Sigma} d(F^*\alpha).$$

Using naturality of the exterior derivative ($d \circ F^* = F^* \circ d$), we get

$$\int_{\partial\Sigma} F^*\alpha = \int_{\Sigma} F^*(d\alpha).$$

Now compute $d\alpha$. Since $d(dx) = d(dy) = 0$,

$$d\alpha = d(P\,dx) + d(Q\,dy) = dP \wedge dx + dQ \wedge dy.$$

Writing $dP = P_x\,dx + P_y\,dy$ and $dQ = Q_x\,dx + Q_y\,dy$, we obtain

$$dP \wedge dx = (P_x\,dx + P_y\,dy) \wedge dx = P_y\,dy \wedge dx = -P_y\,dx \wedge dy,$$

$$dQ \wedge dy = (Q_x\,dx + Q_y\,dy) \wedge dy = Q_x\,dx \wedge dy,$$

hence

$$d\alpha = (Q_x - P_y)\,dx \wedge dy = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx \wedge dy.$$

Write $F = (u, v)$, where $u = x \circ F$ and $v = y \circ F$. Then $F^*(dx) = du$, $F^*(dy) = dv$, and therefore

$$F^*(dx \wedge dy) = du \wedge dv.$$

Also,

$$F^*\alpha = (P \circ F)\,du + (Q \circ F)\,dv.$$

Putting everything together gives the Green/Stokes identity:

$$\int_{\partial\Sigma} \Big((P \circ F)\,du + (Q \circ F)\,dv\Big) = \int_{\Sigma} \Big(\big(\tfrac{\partial Q}{\partial x} - \tfrac{\partial P}{\partial y}\big) \circ F\Big)\,du \wedge dv.$$

When $\Sigma \subset \mathbb{R}^2$ and F is the inclusion map, this reduces to the classical Green's theorem.

Problem 10*. Hairy ball theorem via antipodal map and de Rham cohomology

Statement. For even $n > 2$ there is no nowhere-zero tangent vector field on S^n. Prove this by: (a) showing that a nowhere-zero tangent field on S^n induces a homotopy from id_{S^n} to the antipodal map, and (b) using de Rham cohomology to show that no such homotopy exists.

Solution.

(a) Assume there exists a nowhere-zero tangent vector field V on S^n. Define the normalized field

$$U(x) = \frac{V(x)}{\|V(x)\|} \in T_xS^n.$$

Since $U(x) \in T_xS^n$, we have $U(x) \cdot x = 0$ in $\mathbb{R}^{n+1}$, and $\|U(x)\| = 1$.

Define a map $H : S^n \times [0,1] \to S^n$ by

$$H(x,t) = \cos(\pi t)\,x + \sin(\pi t)\,U(x).$$

Because $x \perp U(x)$ and $\|x\| = \|U(x)\| = 1$,

$$\|H(x,t)\|^2 = \cos^2(\pi t)\|x\|^2 + \sin^2(\pi t)\|U(x)\|^2 = \cos^2(\pi t) + \sin^2(\pi t) = 1,$$

so $H(x,t) \in S^n$ for all (x,t). Moreover,

$$H(x,0) = x \quad \text{and} \quad H(x,1) = \cos(\pi)x + \sin(\pi)U(x) = -x.$$

Thus H is a homotopy from id_{S^n} to the antipodal map $\alpha(x) = -x$.

(b) On S^n we have $H^n_{\mathrm{dR}}(S^n) \cong \mathbb{R}$, generated by the cohomology class $[\omega]$ of any volume form ω with $\int_{S^n} \omega \neq 0$. For a smooth map $f : S^n \to S^n$,

$$f^*([\omega]) = \deg(f)\,[\omega].$$

In particular, homotopic maps induce the same map on de Rham cohomology, hence have the same degree.

Now $\deg(\mathrm{id}_{S^n}) = 1$, while the antipodal map $\alpha(x) = -x$ has degree

$$\deg(\alpha) = (-1)^{n+1}.$$

If n is even, then $(-1)^{n+1} = -1$, so

$$\alpha^*([\omega]) = -[\omega] \neq [\omega] = \mathrm{id}^*([\omega]).$$

Therefore id_{S^n} and α cannot be homotopic when n is even. This contradicts part (a), which produced a homotopy $\mathrm{id}_{S^n} \simeq \alpha$ from a nowhere-zero tangent field.

Hence no nowhere-zero tangent vector field on S^n exists for even $n > 2$.

Examples. For odd spheres $S^{2m+1} \subset \mathbb{C}^{m+1}$, the vector field $V(z) = iz$ is tangent and nowhere zero, so odd spheres admit nowhere-zero tangent vector fields, in contrast to the even-dimensional case.

Problem 1. Local flows and the vector field they generate

Statement. Let U be a flow domain and let $\Phi : U \to X$ be a smooth map satisfying

$$\Phi^s \circ \Phi^t = \Phi^{s+t}$$

whenever this makes sense. Show that Φ is a local flow of a vector field v on X, which you should define.

Solution. Define a vector field v on X by

$$v(x) := \left.\frac{d}{dt}\right|_{t=0} \Phi^t(x) \in T_xX.$$

This is smooth since Φ is smooth in (t,x).

Using $\Phi^{t+h} = \Phi^h \circ \Phi^t$ and $\Phi^0 = \mathrm{id}$ (on the $t = 0$ slice of the flow domain), we compute for fixed x:

$$\frac{\Phi^{t+h}(x) - \Phi^t(x)}{h} = \frac{\Phi^h(\Phi^t(x)) - \Phi^0(\Phi^t(x))}{h}.$$

Letting $h \to 0$ gives

$$\frac{d}{dt}\Phi^t(x) = v(\Phi^t(x)), \qquad \Phi^0(x) = x.$$

Hence Φ is the local flow of v.

Example. On $\mathbb{R}^2$, the rotation flow $\Phi^t(x,y) = (x\cos t - y\sin t,\ x\sin t + y\cos t)$ satisfies $\Phi^s \circ \Phi^t = \Phi^{s+t}$ and its generating vector field is $v(x,y) = (-y,x) = x\,\partial_y - y\,\partial_x$.

Problem 2. Coordinate formulas for Lie derivatives and a planar example

Statement. (a) Compute the coordinate expression for the Lie derivative of a 1-form S_a and of a tensor $T^b{}_a$ along a vector field v. (b) Write down the flow of $v = x\,\partial_y - y\,\partial_x$ on $\mathbb{R}^2$, and compute $\mathcal{L}_v\alpha$ for $\alpha = x\,dy$ directly from the definition.

Solution.

(a) Let $v = v^c\partial_c$ in local coordinates. For a 1-form $S = S_a\,dx^a$,

$$(\mathcal{L}_v S)_a = v^c\,\partial_c S_a + S_c\,\partial_a v^c.$$

For a $(1,1)$-tensor $T = T^b{}_a\,\partial_b \otimes dx^a$,

$$(\mathcal{L}_v T)^b{}_a = v^c\,\partial_c T^b{}_a - T^b{}_c\,\partial_a v^c + T^c{}_a\,\partial_c v^b.$$

(b) The flow solves

$$\dot{x} = -y, \qquad \dot{y} = x,$$

hence

$$\Phi^t(x,y) = (x\cos t - y\sin t,\ x\sin t + y\cos t).$$

Using the definition $\mathcal{L}_v\alpha = \frac{d}{dt}\big|_{t=0}(\Phi^t)^*\alpha$ and $\alpha = x\,dy$, write $(x',y') = \Phi^t(x,y)$ so that $(\Phi^t)^*\alpha = x'\,dy'$. We have

$$x' = x\cos t - y\sin t, \qquad y' = x\sin t + y\cos t, \qquad dy' = \sin t\,dx + \cos t\,dy.$$

Therefore

$$(\Phi^t)^*\alpha = (x\cos t - y\sin t)(\sin t\,dx + \cos t\,dy).$$

Differentiating at $t = 0$ gives

$$\mathcal{L}_v(x\,dy) = x\,dx - y\,dy.$$

Problem 3. Cartan's magic formula

Statement. Prove Cartan's magic formula

$$\mathcal{L}_v\alpha = d(\iota_v\alpha) + \iota_v(d\alpha)$$

for a vector field v and an r-form α on a manifold X, using a local flow $\Phi : U \to X$ of v and the map $F : U \to \mathbb{R}\times X$ given by $F(t,p) = (t,\Phi^t(p))$, viewed as a diffeomorphism onto its image V.

Solution.

(a) On $U \subset \mathbb{R}\times X$ choose coordinates $(t, x^1, \ldots, x^n)$. Any r-form β on U can be written uniquely as

$$\beta = dt \wedge \gamma(t,x) + \eta(t,x),$$

where γ is an $(r-1)$-form in the x-variables and η is an r-form in the x-variables. Then $\iota_{\partial_t}\beta = \gamma$ and $\mathcal{L}_{\partial_t}\beta = dt \wedge \partial_t\gamma + \partial_t\eta$. A direct computation gives

$$d(\iota_{\partial_t}\beta) + \iota_{\partial_t}(d\beta) = dt \wedge \partial_t\gamma + \partial_t\eta = \mathcal{L}_{\partial_t}\beta,$$

so Cartan's formula holds for (U, ∂_t, β).

(b) Lie derivative, contraction, and exterior derivative are natural under diffeomorphisms: for a diffeomorphism F,

$$F^*(\mathcal{L}_W\theta) = \mathcal{L}_{F^*W}(F^*\theta), \quad F^*(\iota_W\theta) = \iota_{F^*W}(F^*\theta), \quad F^*(d\theta) = d(F^*\theta).$$

Moreover $F_*(\partial_t) = \partial_t \oplus v$ on V. Applying part (a) to $F^*(\mathrm{pr}_2^*\alpha)$ and transporting the identity by F yields Cartan's formula on V for $\mathrm{pr}_2^*\alpha$ and $\partial_t \oplus v$.

(c) Let $i_0 : X \to V$ be the inclusion $p \mapsto (0, p)$. Then $i_0^*(\mathrm{pr}_2^*\alpha) = \alpha$. Pulling back the identity from (b) by i_0^* gives

$$\mathcal{L}_v\alpha = d(\iota_v\alpha) + \iota_v(d\alpha),$$

which is Cartan's magic formula on X.

Theory for Problems 1–3 (Flows, Lie derivative, Cartan's formula)

1. Theory: local flows and vector fields

A *(local) flow* on a manifold X is a smooth map

$$\Phi : U \subset \mathbb{R} \times X \to X, \qquad (t, x) \mapsto \Phi^t(x),$$

defined on a *flow domain* U, such that whenever both sides make sense,

$$\Phi^s \circ \Phi^t = \Phi^{s+t},$$

and on the $t = 0$ slice one has

$$\Phi^0(x) = x.$$

This expresses *consistent time evolution*: flowing for time t and then for time s is the same as flowing for time $s + t$.

A vector field is the *infinitesimal velocity rule* for the flow. If a flow Φ is given, its generating vector field v is defined by the initial velocity

$$v(x) = \left.\frac{d}{dt}\right|_{t=0} \Phi^t(x) \in T_xX.$$

Conversely, given a smooth vector field v, standard ODE theory (in local coordinates) produces a local flow Φ whose trajectories satisfy

$$\frac{d}{dt}\Phi^t(x) = v(\Phi^t(x)), \qquad \Phi^0(x) = x.$$

Thus:

$$\text{flow } \Phi \;\longleftrightarrow\; \text{vector field } v,$$

with v describing the infinitesimal motion and Φ describing the finite-time motion.

2. Theory: Lie derivative as "derivative along the flow"

Let v be a vector field on X with local flow Φ^t. The *Lie derivative* measures how a geometric object changes when it is transported along the flow. For a differential form (and more generally any covariant tensor) T, it is defined by

$$\mathcal{L}_vT = \left.\frac{d}{dt}\right|_{t=0} (\Phi^t)^*T.$$

Here $(\Phi^t)^*$ is used so that the comparison is made at the *same point*: $(\Phi^t)^*T$ pulls the value of T at $\Phi^t(x)$ back to x, enabling differentiation at $t = 0$.

In coordinates, $\mathcal{L}_v$ consists of a transport term (directional derivative along v) plus correction terms coming from the variation of the coordinate basis under the flow. This is the origin of the standard index rule:

- each *covariant* index contributes a correction involving derivatives of v with a "minus-type" behavior,
- each *contravariant* index contributes a correction involving derivatives of v with a "plus-type" behavior.

Thus the Lie derivative is best viewed as the infinitesimal action of the diffeomorphism group generated by v.

3. Theory: Cartan's magic formula
For any vector field v and any differential form α on X, Cartan's magic formula states:

$$\boxed{\mathcal{L}_v\alpha \;=\; d(\iota_v\alpha) + \iota_v(d\alpha)}.$$

Here:

- d is the exterior derivative,
- ι_v is interior product (contraction), defined by

$$(\iota_v\alpha)(v_2, \dots, v_r) = \alpha(v, v_2, \dots, v_r).$$

This formula is "magic" because it expresses the Lie derivative—defined via the flow and pullback—purely in terms of two intrinsic operations on forms, d and ι_v. Consequently, it allows computation of $\mathcal{L}_v$ without explicitly writing the flow, and it yields many structural identities (e.g. compatibility with d and Leibniz rules).

A standard proof strategy is to convert "flow on X" into "translation in the $\mathbb{R}$-direction" on $\mathbb{R} \times X$ via the map

$$F(t, p) = (t, \Phi^t(p)).$$

Translation by ∂_t is easy to compute directly, and diffeomorphism-invariance then transfers the identity back to the original vector field v on X.

Problem 4 (Lie derivative, flow invariance, and commuting flows)

Statement. Let v be a vector field on X with local flow Φ^t.

(a) Show that for any tensor T, if $\mathcal{L}_vT = 0$ then $(\Phi^t)^*T = T$ wherever this makes sense.

(b) Let w be another vector field with local flow Ψ^u. For small t, u, show that $\Phi^{-t}\circ\Psi^u\circ\Phi^t$ is the time-u flow of $(\Phi^t)_*w$, and deduce that if $[v, w] = 0$ then Φ^t and Ψ^u commute.

Solution. (a) The Lie derivative is defined by the flow via

$$\frac{d}{dt}\big((\Phi^t)^*T\big) = (\Phi^t)^*(\mathcal{L}_vT)$$

(where defined). If $\mathcal{L}_v T = 0$, then the derivative is 0, so $(\Phi^t)^* T$ is constant in t. At $t = 0$ we have $\Phi^0 = \mathrm{id}$, hence $(\Phi^0)^* T = T$, and therefore $(\Phi^t)^* T = T$.

(b) Fix t and define $F_u = \Phi^{-t} \circ \Psi^u \circ \Phi^t$. Then $F_0 = \mathrm{id}$. Differentiate at $u = 0$:

$$\frac{d}{du}\bigg|_{u=0} F_u(x) = d(\Phi^{-t})_{\Phi^t(x)} \left(\frac{d}{du}\bigg|_{u=0} \Psi^u(\Phi^t(x)) \right) = d(\Phi^{-t})_{\Phi^t(x)}\big(w_{\Phi^t(x)}\big),$$

which is the vector field obtained by pushing w forward by Φ^t, i.e. $(\Phi^t)_* w$ (with this conjugation convention). Hence $u \mapsto F_u$ is the local flow of $(\Phi^t)_* w$.

If $[v, w] = 0$, then $\mathcal{L}_v w = [v, w] = 0$, so w is invariant under Φ^t, i.e. $(\Phi^t)_* w = w$. Therefore $\Phi^{-t} \Psi^u \Phi^t$ is the time-u flow of w, namely Ψ^u itself:

$$\Phi^{-t} \circ \Psi^u \circ \Phi^t = \Psi^u,$$

which implies $\Psi^u \circ \Phi^t = \Phi^t \circ \Psi^u$ (where defined). □

Theory (flows, pullback/pushforward, Lie derivative, bracket)

Flow. A (local) flow of a vector field v on a manifold X is a smooth map $\Phi : \mathcal{D} \subset \mathbb{R} \times X \to X$, $(t, x) \mapsto \Phi^t(x)$ such that

$$\Phi^0 = \mathrm{id}, \qquad \frac{d}{dt}\Phi^t(x) = v_{\Phi^t(x)},$$

and $\Phi^{t+s} = \Phi^t \circ \Phi^s$ whenever both sides are defined.

Pullback and pushforward. If $\varphi : U \to V$ is a diffeomorphism, then for a covariant tensor T the pullback $\varphi^* T$ is defined by composing with $d\varphi$ in each slot, and for a vector field w the pushforward is

$$(\varphi_* w)_{\varphi(x)} := d\varphi_x(w_x).$$

Lie derivative via the flow. For any tensor field T,

$$\mathcal{L}_v T := \frac{d}{dt}\bigg|_{t=0} (\Phi^t)^* T,$$

and one has the identity

$$\frac{d}{dt}\big((\Phi^t)^* T\big) = (\Phi^t)^*(\mathcal{L}_v T).$$

Lie bracket. For vector fields v, w, the Lie bracket is characterized by

$$[v, w](f) = v(w(f)) - w(v(f)) \quad \text{for all smooth } f,$$

and equivalently

$$\mathcal{L}_v w = [v, w].$$

Thus $[v, w] = 0$ means w is invariant under the flow of v and implies the local commutativity of flows.

Problem 5 (Local normal form of a submersion)

Statement. Let X and Y be manifolds of dimensions n and m, and suppose $F : X \to Y$ is a submersion at p. Construct an open neighbourhood U of p and a smooth map $G : U \to \mathbb{R}^{n-m}$ such that $(F|_U, G) : U \to Y \times \mathbb{R}^{n-m}$ is a local diffeomorphism at p. Deduce that there exist local coordinates on X and Y about p and $F(p)$ in which F is given by projection onto the first m components.

Solution. Choose charts $x = (x^1, \dots, x^n) : U_0 \to \mathbb{R}^n$ around p and $y = (y^1, \dots, y^m) : V_0 \to \mathbb{R}^m$ around $F(p)$, and set $f := y \circ F \circ x^{-1} : x(U_0) \subset \mathbb{R}^n \to \mathbb{R}^m$. Since F is a submersion at p, $\operatorname{rank}(Df) = m$ at $x(p)$, so some $m \times m$ minor is invertible. After reordering x^i, assume $\frac{\partial(f^1,\dots,f^m)}{\partial(x^1,\dots,x^m)}(x(p))$ is invertible.

Define $G = (x^{m+1}, \dots, x^n)$ on a smaller neighbourhood $U \subset U_0$. In coordinates let $g(x^1, \dots, x^n) = (x^{m+1}, \dots, x^n)$ and $H = (f, g)$. Then

$$DH(x(p)) = \begin{pmatrix} \frac{\partial f}{\partial(x^1,\dots,x^m)} & \frac{\partial f}{\partial(x^{m+1},\dots,x^n)} \\ 0 & I_{n-m} \end{pmatrix}$$

is invertible, hence by the inverse function theorem H is a local diffeomorphism at $x(p)$. Therefore $(F|_U, G) : U \to Y \times \mathbb{R}^{n-m}$ is a local diffeomorphism at p.

For the normal form, define coordinates on X by $\tilde{x} = (y^1 \circ F, \dots, y^m \circ F,\ x^{m+1}, \dots, x^n)$ and on Y by $\tilde{y} = (y^1, \dots, y^m)$. Then $\tilde{y} \circ F \circ \tilde{x}^{-1}(u^1, \dots, u^n) = (u^1, \dots, u^m)$, i.e. F is projection.

$$\begin{tikzcd} U \subset X \arrow[r, "{(F,G)}"] \arrow[dr, "F"'] & Y \times \mathbb{R}^{n-m} \arrow[d, "\mathrm{pr}_1"] \\ & Y \end{tikzcd}$$

Figure 4: Problem 5: Local normal form of a submersion. The map (F, G) is a local diffeomorphism and $F = \mathrm{pr}_1 \circ (F, G)$.

Object	**Local coordinates**	**Form in coordinates**
Point in X	$(u^1, \dots, u^m,\ u^{m+1}, \dots, u^n)$	—
Point in Y	$(u^1, \dots, u^m)$	—
Submersion F	$(u^1, \dots, u^n) \mapsto (u^1, \dots, u^m)$	$F =$ projection
Complement G	$(u^1, \dots, u^n) \mapsto (u^{m+1}, \dots, u^n)$	$G =$ remaining coords
Combined map (F, G)	$(u^1, \dots, u^n) \mapsto ((u^1, \dots, u^m), (u^{m+1}, \dots, u^n))$	local diffeo

Figure 5: Problem 5: In adapted coordinates, F becomes projection and (F, G) becomes the identity to first order.

Problem 6 (No surjective smooth map to higher dimension)

Statement. Show that there is no surjective smooth map $f : X \to Y$ if $\dim X < \dim Y$.

Solution. Let $\dim X = n$ and $\dim Y = m$ with $n < m$. For every $p \in X$ we have $\operatorname{rank}(df_p) \leq n < m$, hence every p is a critical point and every value in $f(X)$ is a critical value. By Sard's theorem the set of critical values has measure zero in Y (in charts), so $f(X)$ has measure zero and cannot equal Y. Thus f is not surjective.

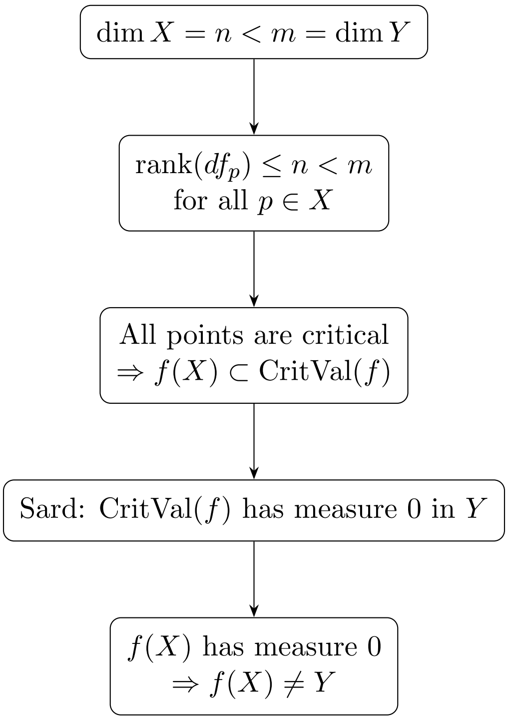


Figure 6: Dimension obstruction to surjectivity via Sard's theorem.

Integrable horizontal distribution D	**Path-independence of transport**
There exist local leaves L with $T_x L = D_x$	Transport depends only on endpoint $\bar{\gamma}(1)$
$\pi\|_L$ is a local diffeo onto a neighbourhood $V \subset Y$	For each q in a fibre-neighbourhood, define $s_q(y) := P_{\bar{\gamma}}(q)$
Local section $s : V \to X$ with image in a leaf	s_q is well-defined (independent of curve)
Local trivialization: neighbourhood $\cong U \times V$	Images of s_q form local leaves tangent to D

Figure 7: Problem 7: Equivalent viewpoints. Integrability produces local leaves; endpoint-only transport produces the same leaves.

Problem 7 (Horizontal lifts, transport, and integrability)

Statement. Let $\pi : X \to Y$ be a submersion, and let D be a k-plane distribution on X transverse to the fibres, where $k = \dim X - \dim Y$. A curve in X is horizontal if it is everywhere tangent to D. Given $p \in X$ and $\bar{\gamma} : [0, 1] \to Y$ with $\bar{\gamma}(0) = \pi(p)$, show that for small $\varepsilon > 0$ there is a unique horizontal curve $\gamma : [0, \varepsilon] \to X$ with $\gamma(0) = p$ and $\pi \circ \gamma = \bar{\gamma}|_{[0,\varepsilon]}$. If $\varepsilon = 1$, define parallel transport by $P_{\bar{\gamma}}(p) = \gamma(1)$. Show that D is integrable iff for all p there exist neighbourhoods $U \subset \pi^{-1}(\pi(p))$ and $V \subset Y$ such that for all $q \in U$ and all curves $\bar{\gamma}$ in V with $\bar{\gamma}(0) = \pi(q)$, the transport $P_{\bar{\gamma}}(q)$ exists and depends only on q and $\bar{\gamma}(1)$.

Solution. (Existence/uniqueness of horizontal lift.) Transversality means $T_xX = D_x \oplus \ker(d\pi_x)$, hence $d\pi_x|_{D_x} : D_x \to T_{\pi(x)}Y$ is an isomorphism near p. Given $\bar{\gamma}$, define a time-dependent vector field W_t near p by $W_t(x) \in D_x$ and $d\pi_x(W_t(x)) = \dot{\bar{\gamma}}(t)$. Solve the ODE $\dot{\gamma}(t) = W_t(\gamma(t))$, $\gamma(0) = p$. ODE theory gives a unique solution on $[0, \varepsilon]$. By construction $\dot{\gamma}(t) \in D_{\gamma(t)}$ and $\pi \circ \gamma = \bar{\gamma}$ on $[0, \varepsilon]$.

(Integrable $\Rightarrow$ endpoint dependence.) If D is integrable, let L_q be the leaf through q. Since $d\pi|_D$ is an isomorphism, $\pi|_{L_q}$ is a local diffeomorphism onto some V, so it has a local inverse $s_q : V \to X$ with $\pi \circ s_q = \mathrm{id}$ and $s_q(\pi(q)) = q$. Then for any curve $\bar{\gamma}$ in V starting at $\pi(q)$, the lift is $\gamma(t) = s_q(\bar{\gamma}(t))$, so $P_{\bar{\gamma}}(q) = s_q(\bar{\gamma}(1))$, depending only on the endpoint.

(Endpoint dependence $\Rightarrow$ integrable.) Assume the stated property. For $q \in U$, define $s_q(y)$ for $y \in V$ as the transport of q along any curve in V from $\pi(q)$ to y. Path-independence makes s_q well-defined. For any curve $\bar{\gamma}$ in V, $t \mapsto s_q(\bar{\gamma}(t))$ is the horizontal lift, hence $(ds_q)_y(T_yY) \subset D_{s_q(y)}$. By dimension, $(ds_q)_y(T_yY) = D_{s_q(y)}$, so $\mathrm{im}(s_q)$ is an integral manifold of D. These images give a local foliation, hence D is integrable.

$$\begin{tikzcd}
\pi^{-1}(\bar{\gamma}(0)) \ni p \arrow[r, "P_{\bar{\gamma}}"] \arrow[d, "\pi"'] & \pi^{-1}(\bar{\gamma}(1)) \ni \gamma(1) \arrow[d, "\pi"] \\
\bar{\gamma}(0) \arrow[r, "\bar{\gamma}"'] & \bar{\gamma}(1)
\end{tikzcd}$$

Figure 8: Problem 7: Parallel transport along $\bar{\gamma}$ sends p to $\gamma(1)$, where γ is the horizontal lift.

Theory for Problem 7 (Ehresmann connection viewpoint)

Submersion and vertical bundle. A smooth map $\pi : X \to Y$ is a *submersion* if $d\pi_x : T_xX \to T_{\pi(x)}Y$ is surjective for all x. The *vertical subspace* is $V_x := \ker(d\pi_x)$, tangent to the fibre $\pi^{-1}(\pi(x))$.

Horizontal distribution (connection). A distribution $D \subset TX$ is *transverse to the fibres* if

$$T_xX = V_x \oplus D_x \quad \text{for all } x,$$

so $d\pi_x|_{D_x} : D_x \to T_{\pi(x)}Y$ is an isomorphism. Such a D is an Ehresmann connection.

Horizontal curves, lifts, transport. A curve γ in X is *horizontal* if $\dot{\gamma}(t) \in D_{\gamma(t)}$. Given $\bar{\gamma}$ in Y and $p \in \pi^{-1}(\bar{\gamma}(0))$, a *horizontal lift* is γ with $\pi \circ \gamma = \bar{\gamma}$ and $\gamma(0) = p$. If defined to time 1, $\gamma(1)$ is the *parallel transport* of p along $\bar{\gamma}$.

Integrability (Frobenius). D is *integrable* if through each point there is an immersed leaf tangent to D. Equivalently (Frobenius), D is integrable iff it is involutive: $[X, Y] \in \Gamma(D)$ for all $X, Y \in \Gamma(D)$.

Problem 8 (Orthogonal and special unitary groups as embedded Lie subgroups)

Statement.

(a) By considering the map $F : GL(n,\mathbb{R}) \to \{\text{symmetric matrices}\}$ given by $F(A) = A^T A$, show that $O(n)$ is an embedded Lie subgroup of $GL(n,\mathbb{R})$. Identify $\mathfrak{o}(n)$ as a subspace of $\mathfrak{gl}(n,\mathbb{R})$.

(b) Show that $SU(n)$ is a Lie subgroup of $GL(n,\mathbb{C})$ and similarly identify its Lie algebra.

Solution. (a) Let $\mathrm{Sym}(n) = \{S \in M_n(\mathbb{R}) : S^T = S\}$ and define $F(A) = A^T A$. Then

$$O(n) = \{A \in GL(n,\mathbb{R}) : A^T A = I\} = F^{-1}(I).$$

Identify $T_A GL(n,\mathbb{R}) \cong M_n(\mathbb{R})$. For $B \in M_n(\mathbb{R})$,

$$(dF)_A(B) = \frac{d}{dt}\Big|_0 (A+tB)^T(A+tB) = B^T A + A^T B \in \mathrm{Sym}(n).$$

If $A \in O(n)$ and $S \in \mathrm{Sym}(n)$ is arbitrary, choose $B = \frac{1}{2}AS$. Then

$$(dF)_A(B) = \tfrac{1}{2}S^T(A^T A) + \tfrac{1}{2}(A^T A)S = \tfrac{1}{2}S^T + \tfrac{1}{2}S = S,$$

so $(dF)_A$ is surjective. Hence I is a regular value of F and $O(n) = F^{-1}(I)$ is an embedded submanifold of $GL(n,\mathbb{R})$. Since it is a subgroup, it is an embedded Lie subgroup.

Its Lie algebra is $T_I O(n)$. Differentiating $A^T A = I$ at $A = I$ gives

$$0 = \frac{d}{dt}\Big|_0 (A(t)^T A(t)) = X^T + X,$$

so

$$\mathfrak{o}(n) = \{X \in \mathfrak{gl}(n,\mathbb{R}) : X^T + X = 0\},$$

the skew-symmetric matrices.

(b) Let $U(n) = \{A \in GL(n,\mathbb{C}) : A^*A = I\}$ and $SU(n) = \{A \in U(n) : \det A = 1\}$. As in (a), the map $F_{\mathbb{C}}(A) = A^*A$ has $U(n) = F_{\mathbb{C}}^{-1}(I)$, hence $U(n)$ is an embedded Lie subgroup.

The determinant $\det : U(n) \to S^1$ is smooth and $SU(n) = \det^{-1}(1)$. Its differential at I satisfies

$$\frac{d}{dt}\Big|_0 \det(I + tX) = \mathrm{tr}(X).$$

For $X \in \mathfrak{u}(n) = T_I U(n) = \{X : X^* + X = 0\}$, $\mathrm{tr}(X) \in i\mathbb{R} \cong T_1 S^1$, and the map $X \mapsto \mathrm{tr}(X)$ is surjective (e.g. $X = i\lambda I$). Thus 1 is a regular value and $SU(n)$ is an embedded Lie subgroup of $U(n) \subset GL(n,\mathbb{C})$.

Finally,

$$\mathfrak{su}(n) = \{X \in \mathfrak{gl}(n,\mathbb{C}) : X^* + X = 0,\ \mathrm{tr}(X) = 0\}.$$

Theory for Problem 8 (matrix Lie groups and Lie algebras)

Matrix Lie groups. A matrix Lie group is a subgroup $G \subset GL(N, \mathbb{F})$ that is an embedded submanifold.

Lie algebra. The Lie algebra is $\mathfrak{g} = T_I G \subset M_N(\mathbb{F})$, with bracket $[X, Y] = XY - YX$. Equivalently,

$$\mathfrak{g} = \{X \in M_N(\mathbb{F}) : \exp(tX) \in G \text{ for all small } t\}.$$

Regular level sets. If $F : GL(N, \mathbb{F}) \to W$ is smooth and c is a regular value, then $F^{-1}(c)$ is an embedded submanifold.

Standard examples.

$$O(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I\}, \quad \mathfrak{o}(n) = \{X : X^T + X = 0\}.$$

$$U(n) = \{A \in GL(n, \mathbb{C}) : A^* A = I\}, \quad \mathfrak{u}(n) = \{X : X^* + X = 0\}.$$

$$SU(n) = \{A \in U(n) : \det A = 1\}, \quad \mathfrak{su}(n) = \{X \in \mathfrak{u}(n) : \operatorname{tr} X = 0\}.$$

Group	**Defining constraint**	**Lie algebra condition at I**
$O(n)$	$A^T A = I$	$X^T + X = 0$ (skew-symmetric)
$U(n)$	$A^* A = I$	$X^* + X = 0$ (skew-Hermitian)
$SU(n)$	$A^* A = I$, $\det A = 1$	$X^* + X = 0$, $\operatorname{tr}(X) = 0$

Figure 9: Problem 8: Matrix Lie groups as constraint sets and their Lie algebras from differentiation at the identity.

Step	**What to write**
Define F	$F : GL(n, \mathbb{R}) \to \mathrm{Sym}(n), \quad F(A) = A^T A$
Level set	$O(n) = F^{-1}(I)$
Differentiate	$(dF)_I(X) = X^T + X$
Kernel	$T_I O(n) = \ker(dF)_I = \{X : X^T + X = 0\} = \mathfrak{o}(n)$
Analog for $SU(n)$	Use $F(A) = A^* A$ plus $\det(A) = 1$; get $X^* + X = 0$ and $\operatorname{tr}(X) = 0$

Figure 10: Problem 8: Proof skeleton chart (regular level set + tangent space computation).

Problem 9 (Differentials of multiplication and inversion; sign in the right-invariant bracket)

Statement. For a Lie group G compute the derivatives $D_{(e,e)} m : \mathfrak{g} \oplus \mathfrak{g} \to \mathfrak{g}$ and $D_e i : \mathfrak{g} \to \mathfrak{g}$ of the multiplication and inversion maps m and i at the identity. Show that i_* exchanges left-invariant and right-invariant vector fields, and deduce that the bracket operation on $\mathfrak{g}$ defined using right-invariant (rather than left-invariant) vector fields differs from the usual one by a sign.

Solution. Let $m(g, h) = gh$ and $i(g) = g^{-1}$.

Prob.	**Main tool**	**Main takeaway**
5	Inverse Function Theorem	Submersion $\Rightarrow$ local product coordinates, F becomes projection
6	Sard's Theorem	If $\dim X < \dim Y$, image has measure 0 $\Rightarrow$ no surjective smooth map
7	ODE + Frobenius idea	Horizontal lifts exist/unique; integrability $\Leftrightarrow$ endpoint-only transport
8	Regular level set + tangent space	$O(n), SU(n)$ are embedded Lie subgroups; compute Lie algebras by differentiating constraints

Figure 11: Problems 5–8: Tools and conclusions.

Multiplication at (e,e). Take curves $g(t), h(t)$ with $g(0) = h(0) = e$ and $g'(0) = X$, $h'(0) = Y$. Then

$$D_{(e,e)}m(X,Y) = \frac{d}{dt}\Big|_0 g(t)h(t) = X + Y.$$

Inversion at e. Using $g(t) = \exp(tX)$ gives $i(g(t)) = \exp(-tX)$, hence

$$D_e i(X) = \frac{d}{dt}\Big|_0 \exp(-tX) = -X.$$

i_* exchanges left- and right-invariant fields. For $X \in \mathfrak{g}$ define $X^L(g) = (dL_g)_e X$ and $X^R(g) = (dR_g)_e X$. The identity

$$i \circ L_g = R_{g^{-1}} \circ i$$

implies, upon differentiation at e,

$$(di)_g \circ (dL_g)_e = (dR_{g^{-1}})_e \circ (di)_e = -(dR_{g^{-1}})_e.$$

Thus, at $g^{-1} = i(g)$,

$$(i_* X^L)_{g^{-1}} = (di)_g(X^L_g) = -(dR_{g^{-1}})_e X = -(X^R)_{g^{-1}}.$$

So inversion sends left-invariant fields to (minus) right-invariant ones (and similarly in the other direction).

Sign in the right-invariant bracket. Pushforwards preserve Lie brackets: $i_*[U,V] = [i_*U, i_*V]$. Apply this to $U = X^L$, $V = Y^L$ and evaluate at e:

$$(di)_e([X^L, Y^L]_e) = [i_*X^L, i_*Y^L]_e = [-X^R, -Y^R]_e = [X^R, Y^R]_e.$$

Since $(di)_e = -\mathrm{Id}$, we get

$$[X^R, Y^R]_e = -[X^L, Y^L]_e.$$

Hence defining $[X,Y]_{\text{right}} := [X^R, Y^R]_e$ yields

$$[X,Y]_{\text{right}} = -[X,Y]_{\text{usual}}.$$

Theory for Problem 9 (Lie group actions, orbits, quotients)

Smooth action. A smooth (left) action of a Lie group G on a manifold M is a smooth map $\alpha : G \times M \to M$, $(g,p) \mapsto g \cdot p$ with $e \cdot p = p$ and $g \cdot (h \cdot p) = (gh) \cdot p$.

Orbit and stabilizer. The orbit of p is $\mathcal{O}_p = G \cdot p = \{g \cdot p : g \in G\}$. The stabilizer is $G_p = \{g \in G : g \cdot p = p\}$, a Lie subgroup.

Infinitesimal action. For $X \in \mathfrak{g} = T_eG$ define the fundamental vector field

$$X_M(p) = \frac{d}{dt}\Big|_0 \big(\exp(tX) \cdot p\big).$$

Then $T_p\mathcal{O}_p = \{X_M(p) : X \in \mathfrak{g}\}$ and (up to sign convention) $[X_M, Y_M] = \pm[X,Y]_M$.

Quotients. If the action is free and proper, then M/G is a smooth manifold and the projection $\pi : M \to M/G$ is a submersion; $M \to M/G$ is a principal G-bundle.

Problem 10 (Morphisms intertwine exponentials and induce Lie algebra homomorphisms)

Statement. Let $F : H \to G$ be a morphism of Lie groups, i.e. a smooth map which is a group homomorphism. Show that

$$F(\exp_H(\xi)) = \exp_G(D_eF(\xi)) \quad \text{for all } \xi \in \mathfrak{h},$$

and deduce that D_eF is a Lie algebra homomorphism, i.e. a linear map which respects the bracket operation. This shows, in particular, that if H is an embedded Lie subgroup of G then the exponential map and bracket on $\mathfrak{h}$ are the restrictions of those on $\mathfrak{g}$.

Solution. Fix $\xi \in \mathfrak{h}$. The curve $\gamma(t) = \exp_H(t\xi)$ is a one-parameter subgroup of H. Since F is a group homomorphism, $F \circ \gamma$ is a one-parameter subgroup of G. Its derivative at 0 is

$$\frac{d}{dt}\Big|_0 F(\exp_H(t\xi)) = (D_eF)(\xi) \in \mathfrak{g}.$$

A one-parameter subgroup in G is uniquely determined by its tangent at $t = 0$, and equals $t \mapsto \exp_G(tX)$ where X is that tangent. Therefore for all t,

$$F(\exp_H(t\xi)) = \exp_G(t\, D_eF(\xi)).$$

Setting $t = 1$ gives $F(\exp_H(\xi)) = \exp_G(D_eF(\xi))$.

To show D_eF preserves brackets, let $\xi, \eta \in \mathfrak{h}$ and let ξ^L, η^L be the corresponding left-invariant vector fields on H. The identity $F \circ L_h = L_{F(h)} \circ F$ implies

$$F_*(\xi^L) = (D_eF(\xi))^L, \qquad F_*(\eta^L) = (D_eF(\eta))^L.$$

Naturality of brackets under pushforward gives

$$F_*[\xi^L, \eta^L] = [F_*\xi^L, F_*\eta^L].$$

Evaluating at e yields

$$D_eF([\xi,\eta]) = [D_eF(\xi), D_eF(\eta)],$$

so $D_eF : \mathfrak{h} \to \mathfrak{g}$ is a Lie algebra homomorphism.

If $H \subset G$ is an embedded Lie subgroup and F is the inclusion, then $\exp_H$ and the bracket on $\mathfrak{h}$ are the restrictions of $\exp_G$ and the bracket on $\mathfrak{g}$.

Theory for Problem 10 (left-invariant fields, exponential, Ad)

Left-invariant fields and Lie algebra. A vector field X on G is left-invariant if $(dL_g)_h(X_h) = X_{gh}$ for all g, h. Evaluation at e identifies left-invariant fields with $\mathfrak{g} = T_eG$. The bracket of left-invariant fields induces the Lie bracket on $\mathfrak{g}$.

Exponential map. For each $X \in \mathfrak{g}$ there is a unique one-parameter subgroup $\gamma_X : \mathbb{R} \to G$ with $\gamma_X'(0) = X$. Define $\exp(X) = \gamma_X(1)$. Then $\exp(tX) = \gamma_X(t)$ and $d(\exp)_0 = \mathrm{id}$.

Adjoint representation. Conjugation $C_g(h) = ghg^{-1}$ gives $\mathrm{Ad}_g = (dC_g)_e : \mathfrak{g} \to \mathfrak{g}$. Differentiating yields $\mathrm{ad}_X(Y) = \frac{d}{dt}\big|_0 \mathrm{Ad}_{\exp(tX)}Y$, and $\mathrm{ad}_X(Y) = [X, Y]$. Moreover $g\exp(X)g^{-1} = \exp(\mathrm{Ad}_g X)$.

Problem 11 (Coset foliations and integrability of left-invariant distributions)

Statement. Let G be a Lie group. Given an embedded Lie subgroup H, show that its left cosets induce a foliation of G, with tangent distribution $\{\xi^L : \xi \in \mathfrak{h}\}$. Given instead a subspace $\mathfrak{h}$ of $\mathfrak{g}$, show that the distribution $\{\xi^L : \xi \in \mathfrak{h}\}$ on G arises from a foliation iff $\mathfrak{h}$ is actually a Lie subalgebra of $\mathfrak{g}$, i.e. $\mathfrak{h}$ is closed under the Lie bracket on $\mathfrak{g}$. If this holds, must $\mathfrak{h}$ be the Lie algebra of an embedded subgroup of G?

Solution. *Cosets give a foliation.* For each $g \in G$, the left coset gH is the image of H under the diffeomorphism L_g, hence an embedded submanifold. The cosets partition G and locally define a foliation (equivalently, the quotient map $G \to G/H$ is a submersion when H is embedded). At $gh \in gH$,

$$T_{gh}(gH) = (dL_g)_h(T_hH) = (dL_{gh})_e(\mathfrak{h}) = \{\xi^L_{gh} : \xi \in \mathfrak{h}\}.$$

Thus the tangent distribution is $\mathcal{D}_p = \{\xi^L_p : \xi \in \mathfrak{h}\}$.

Integrability $\Longleftrightarrow$ $\mathfrak{h}$ is a Lie subalgebra. Given a vector subspace $\mathfrak{h} \subset \mathfrak{g}$, define a left-invariant distribution

$$\mathcal{D}_g := (dL_g)_e(\mathfrak{h}) = \{\xi^L_g : \xi \in \mathfrak{h}\}.$$

By Frobenius, $\mathcal{D}$ arises from a foliation iff it is involutive. Since $\mathcal{D}$ is spanned by left-invariant fields ξ^L with $\xi \in \mathfrak{h}$, involutivity is equivalent to

$$[\xi^L, \eta^L] \in \Gamma(\mathcal{D}) \quad \text{for all } \xi, \eta \in \mathfrak{h}.$$

But $[\xi^L, \eta^L] = [\xi, \eta]^L$, so this holds iff $[\xi, \eta] \in \mathfrak{h}$ for all $\xi, \eta \in \mathfrak{h}$, i.e. iff $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$.

Embeddedness need not hold. If $\mathfrak{h}$ is a Lie subalgebra, it integrates to a unique connected *immersed* Lie subgroup with Lie algebra $\mathfrak{h}$, but it need not be embedded (equivalently, it need not be closed in G). For example, in $G = \mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$ take $\mathfrak{h} = \mathrm{span}\{(1, \alpha)\}$ with $\alpha \notin \mathbb{Q}$. The corresponding connected subgroup is dense in $\mathbb{T}^2$, hence not an embedded submanifold. Thus $\mathfrak{h}$ need not be the Lie algebra of an embedded subgroup unless the integral subgroup is closed.

Theory for Problem 11 (Maurer–Cartan form and structure equations)

Left translation. For a Lie group G, left translation $L_g(h) = gh$ induces isomorphisms

$$(dL_g)_e : \mathfrak{g} = T_eG \xrightarrow{\cong} T_gG.$$

Hence $(dL_{g^{-1}})_g : T_gG \to \mathfrak{g}$ identifies each tangent space with the Lie algebra.

Maurer–Cartan form. The (left) Maurer–Cartan form is the $\mathfrak{g}$-valued 1-form

$$\theta \in \Omega^1(G;\mathfrak{g}), \qquad \theta_g := (dL_{g^{-1}})_g : T_gG \to \mathfrak{g}.$$

It satisfies $(L_h)^*\theta = \theta$ and $\theta_e = \mathrm{id}_{\mathfrak{g}}$. For matrix groups, $\theta = g^{-1}dg$.

Left-invariant 1-forms. A 1-form α is left-invariant if $(L_h)^*\alpha = \alpha$ for all h. Left-invariant 1-forms correspond to $\mathfrak{g}^*$ via $\alpha_g(v) = \lambda(\theta_g(v))$ for $\lambda \in \mathfrak{g}^*$.

Structure constants and components. Choose a basis $\{e_i\}$ of $\mathfrak{g}$ and dual left-invariant forms $\{\omega^i\}$. If $[e_i, e_j] = \sum_k c_{ij}^k e_k$, then

$$d\omega^k = -\frac{1}{2}\sum_{i,j} c_{ij}^k\, \omega^i \wedge \omega^j.$$

Maurer–Cartan equation. In $\mathfrak{g}$-valued form,

$$d\theta + \tfrac{1}{2}[\theta,\theta] = 0.$$

Equivalently, for left-invariant vector fields X, Y,

$$d\omega(X,Y) = -\omega([X,Y]).$$

Right translations and Ad. Often one uses $(R_h)^*\theta = \mathrm{Ad}_{h^{-1}}\theta$ to relate left/right invariance.

Problem 1 — Properness of standard Lie group actions

Exact statement. Show that the following Lie group actions are proper.

1. The action of $\mathbb{K}^\times$ on $\mathbb{K}^{n+1} \setminus \{0\}$ by rescaling, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$.
2. The right- (or left-) translation action of an embedded Lie subgroup $H \subset G$ on G.
3. Any action of a compact group.

Solution. Recall: an action $G \times M \to M$ is proper if the map

$$\Phi : G \times M \longrightarrow M \times M, \qquad (g,x) \longmapsto (g\cdot x,\ x)$$

is a proper map (preimages of compact sets are compact). On manifolds, this is equivalent to the sequential criterion:

$$\Phi(g_i, x_i) \to (y,x) \quad\Longrightarrow\quad (g_i, x_i) \text{ has a convergent subsequence.}$$

(a) Rescaling action of $\mathbb{K}^\times$ on $\mathbb{K}^{n+1} \setminus \{0\}$. Let

$$(\lambda_i, x_i) \in \mathbb{K}^\times \times \big(\mathbb{K}^{n+1} \setminus \{0\}\big)$$

be such that

$$(\lambda_i x_i,\ x_i) \longrightarrow (y,x) \qquad \text{in} \qquad \big(\mathbb{K}^{n+1} \setminus \{0\}\big) \times \big(\mathbb{K}^{n+1} \setminus \{0\}\big).$$

Then

$$x_i \to x \neq 0.$$

Choose an index j such that $x^{(j)} \neq 0$. For i large, $x_i^{(j)} \neq 0$, and

$$\lambda_i = \frac{(\lambda_i x_i)^{(j)}}{x_i^{(j)}} \longrightarrow \frac{y^{(j)}}{x^{(j)}} =: \lambda \in \mathbb{K}^\times.$$

Hence

$$\lambda_i \to \lambda \qquad \text{and} \qquad x_i \to x,$$

so

$$(\lambda_i, x_i) \to (\lambda, x).$$

Therefore the action is proper.

(b) Translation action of an embedded Lie subgroup $H \subset G$ on G. Consider the right translation action

$$H \times G \longrightarrow G, \qquad (h, g) \longmapsto gh.$$

Assume

$$(g_i h_i,\ g_i) \longrightarrow (g', g) \qquad \text{in} \qquad G \times G.$$

Then

$$g_i \to g \qquad \text{and} \qquad g_i h_i \to g',$$

so

$$h_i = g_i^{-1}(g_i h_i) \longrightarrow g^{-1} g' =: h.$$

Since H is embedded (in particular, closed in G), the limit satisfies

$$h \in H.$$

Hence (h_i, g_i) converges (after passing to a subsequence if desired), so the action is proper. (The left action is analogous.)

(c) Any action of a compact group. Let a compact Lie group K act on M. Suppose

$$(k_i \cdot x_i,\ x_i) \longrightarrow (y, x).$$

Then

$$x_i \to x.$$

By compactness, (k_i) has a convergent subsequence

$$k_{i_m} \to k \in K.$$

Thus

$$(k_{i_m}, x_{i_m}) \to (k, x),$$

so the action is proper.

Problem 2 — From local connection forms to a global connection (and back)

Exact statement.

(a) Given a vector bundle $\pi : E \to B$ and a collection of local $\mathfrak{gl}(k, \mathbb{R})$-valued 1-forms A_α satisfying the preliminary definition of a connection, show that the $\mathfrak{gl}(k, \mathbb{R})$-valued 1-form A constructed from the A_α is well-defined (consistent on overlaps) and satisfies the two conditions on a connection.

Hint: Show that $\quad D_\beta f_\alpha = (R_{g_{\beta\alpha}^{-1}})_* D_\alpha f_\alpha + f_\beta(b) \cdot \eta, \qquad \eta = g_{\beta\alpha}\, dg_{\beta\alpha}^{-1} \in \mathfrak{gl}(k, \mathbb{R}).$

(b) Conversely show that if A is a connection then the local forms $f_\alpha^* A$ satisfy the preliminary definition.

Solution. Let $P = \mathrm{Fr}(E)$ be the principal bundle of frames, with structure group

$$G = \mathrm{GL}(k, \mathbb{R}).$$

Let $\{U_\alpha\}$ be a trivializing cover with local sections (local frames)

$$f_\alpha : U_\alpha \longrightarrow P.$$

On overlaps $U_{\alpha\beta} = U_\alpha \cap U_\beta$, there are transition functions

$$g_{\beta\alpha} : U_{\alpha\beta} \longrightarrow G$$

such that

$$f_\alpha(b) = f_\beta(b)\, g_{\beta\alpha}(b).$$

(a) Well-definedness and the connection axioms. Assume the preliminary definition, i.e. the local 1-forms satisfy on overlaps

$$A_\alpha = g_{\beta\alpha}\, A_\beta\, g_{\beta\alpha}^{-1} + g_{\beta\alpha}\, dg_{\beta\alpha}^{-1}.$$

For $p \in P|_{U_\alpha}$ write uniquely

$$p = f_\alpha(\pi(p)) \cdot g_\alpha(p), \qquad g_\alpha(p) \in G.$$

Define a $\mathfrak{gl}(k, \mathbb{R})$-valued 1-form on $P|_{U_\alpha}$ by

$$A = \mathrm{Ad}_{g_\alpha(p)^{-1}}\big(\pi^* A_\alpha\big) + g_\alpha^{-1} dg_\alpha.$$

On an overlap, the factorization satisfies

$$g_\beta(p) = g_{\beta\alpha}(\pi(p))\, g_\alpha(p).$$

Using the overlap rule for A_α and the product rule for Maurer–Cartan forms,

$$(g_{\beta\alpha} g_\alpha)^{-1} d(g_{\beta\alpha} g_\alpha) = g_\alpha^{-1}(g_{\beta\alpha}^{-1} dg_{\beta\alpha}) g_\alpha + g_\alpha^{-1} dg_\alpha,$$

one checks that the expression for A obtained from α equals the one obtained from β. Hence A is globally well-defined.

Moreover, A satisfies the two connection axioms:

$$R_h^* A = \mathrm{Ad}_{h^{-1}} A \qquad \text{for all} \qquad h \in G,$$

and for every $\xi \in \mathfrak{g}$,

$$A(\xi^\#) = \xi.$$

(b) Conversely, local forms from a global connection. Let A be a connection form on P and set

$$A_\alpha := f_\alpha^* A.$$

On overlaps,

$$f_\alpha = f_\beta \cdot g_{\beta\alpha},$$

and using equivariance of A together with the differential contribution from $dg_{\beta\alpha}$ (the hint term), one gets

$$A_\alpha = g_{\beta\alpha}\, A_\beta\, g_{\beta\alpha}^{-1} + g_{\beta\alpha}\, dg_{\beta\alpha}^{-1}.$$

Thus the $f_\alpha^* A$ satisfy the preliminary definition.

Problem 3 — Covariant exterior derivative: Leibniz, uniqueness, and curvature identity

Exact statement. Let A be a connection on a vector bundle E.

(a) Prove the Leibniz rule

$$d^A(fs) = f\, d^A s + s \otimes df$$

for sections s and functions f.

(b) Conversely, show that every $\mathbb{R}$-linear map

$$D : \Gamma(E) \longrightarrow \Omega^1(B; E)$$

satisfying

$$D(fs) = f\, Ds + s \otimes df$$

is given by d^A for a unique connection A on E.

(c) Show that for any E-valued p-form σ,

$$(d^A)^2 \sigma = F \wedge \sigma.$$

Solution. Let ∇ be the covariant derivative associated to A, and define d^A on sections by

$$(d^A s)(X) = \nabla_X s.$$

(a) For any vector field X,

$$(d^A(fs))(X) = \nabla_X(fs) = X(f)\, s + f\, \nabla_X s.$$

Therefore

$$d^A(fs) = f\, d^A s + s \otimes df.$$

(b) Given D satisfying the Leibniz rule, define

$$\nabla_X s := (Ds)(X).$$

Then for any function f,

$$\nabla_X(fs) = (D(fs))(X) = \big(fDs + s \otimes df\big)(X) = f(Ds)(X) + X(f)s = f\nabla_X s + X(f)s.$$

Thus ∇ is a connection and, by definition,

$$Ds = d^A s.$$

Uniqueness follows since D determines all values $\nabla_X s$.

(c) In a local trivialization one has

$$d^A = d + A \wedge (\cdot),$$

and the curvature is

$$F = dA + A \wedge A.$$

Hence

$$(d^A)^2 = (d + A\wedge)^2 = (dA + A \wedge A) \wedge (\cdot) = F \wedge (\cdot),$$

so for any E-valued p-form σ,

$$(d^A)^2\sigma = F \wedge \sigma.$$

Problem 4 — Curvature as vertical commutator and small-square holonomy

Exact statement. Fix a G-bundle $\pi : P \to B$ with a connection A.

(a) Given vector fields v, w on B, let $\hat{v}, \hat{w}$ denote their (unique) horizontal lifts to P. Show that the vertical component of $[\hat{v}, \hat{w}]$ at a point p is

$$p \cdot \big(-F(\hat{v}, \hat{w})\big).$$

(b) Take local coordinates on B around $\pi(p)$, and define $\gamma(t)$ to be the result of parallel transporting p for time t in the x^i-direction, then time t in the x^j-direction, then the other two sides of the square. Show that

$$\dot{\gamma}(0) = 0 \qquad \text{and} \qquad \ddot{\gamma}(0) = p \cdot \big(-2F(u_i, u_j)\big),$$

where u_i and u_j are any lifts of ∂_{x^i} and ∂_{x^j} to p.

Solution. Let $A \in \Omega^1(P, \mathfrak{g})$ be the connection form and let the curvature be

$$F = dA + \tfrac{1}{2}[A \wedge A].$$

(a) Since $\hat{v}$ and $\hat{w}$ are horizontal,

$$A(\hat{v}) = 0 \qquad \text{and} \qquad A(\hat{w}) = 0.$$

Thus

$$F(\hat{v}, \hat{w}) = dA(\hat{v}, \hat{w}) = \hat{v}\big(A(\hat{w})\big) - \hat{w}\big(A(\hat{v})\big) - A([\hat{v}, \hat{w}]) = -A([\hat{v}, \hat{w}]).$$

Hence
$$A([\hat{v},\hat{w}]) = -F(\hat{v},\hat{w}).$$
The vertical component of a tangent vector $X \in T_pP$ is the fundamental vertical vector
$$\big(A(X)\big)^{\#}_p,$$
so
$$\big([\hat{v},\hat{w}]\big)^{\text{vert}}_p = \big(A([\hat{v},\hat{w}])\big)^{\#}_p = \big(-F(\hat{v},\hat{w})\big)^{\#}_p = p\cdot\big(-F(\hat{v},\hat{w})\big).$$

(b) Let u_i, u_j be the horizontal lifts of $\partial_{x^i}, \partial_{x^j}$ at p, and let Φ_i^t, Φ_j^t be their flows. The parallel-transport around the coordinate square can be written as the commutator of flows
$$\gamma(t) = \Phi_j^{-t}\circ\Phi_i^{-t}\circ\Phi_j^{t}\circ\Phi_i^{t}(p).$$
Standard flow-commutator calculus gives
$$\dot\gamma(0) = 0 \qquad \text{and} \qquad \ddot\gamma(0) = 2\,[u_i,u_j]_p.$$
Moreover,
$$\pi_*[u_i,u_j] = [\pi_*u_i, \pi_*u_j] = [\partial_{x^i},\partial_{x^j}] = 0,$$
so $[u_i,u_j]_p$ is vertical. By part (a),
$$[u_i,u_j]_p = p\cdot\big(-F(u_i,u_j)\big).$$
Therefore
$$\ddot\gamma(0) = 2[u_i,u_j]_p = p\cdot\big(-2F(u_i,u_j)\big).$$

Problem 5 (Hopf bundle: local connection and curvature on U_0)

Statement. Recall the connection we defined on the Hopf bundle
$$S^{2n+1} \longrightarrow \mathbb{CP}^n$$
via its horizontal distribution
$$H_p \;=\; T_pS^{2n+1}\cap i\cdot T_pS^{2n+1}.$$
Trivialise the bundle over $U_0 \subset \mathbb{CP}^n$, and compute the local connection 1-form A and curvature F in this trivialisation.

Solution. Take the standard affine chart
$$U_0 \;=\; \{[z_0:\cdots:z_n]\in\mathbb{CP}^n \;:\; z_0\neq 0\},$$
with affine coordinates
$$w_j \;=\; \frac{z_j}{z_0} \qquad (j=1,\dots,n).$$
Define a local section $s_0 : U_0 \to S^{2n+1}\subset\mathbb{C}^{n+1}$ by
$$s_0(w) \;=\; \frac{(1,w_1,\dots,w_n)}{\sqrt{1+\|w\|^2}},$$

where

$$\|w\|^2 = \sum_{j=1}^{n} |w_j|^2.$$

This yields a trivialisation

$$U_0 \times S^1 \longrightarrow \pi^{-1}(U_0), \qquad (w, e^{i\theta}) \longmapsto e^{i\theta}\, s_0(w).$$

On S^{2n+1} take the standard Hopf connection 1-form

$$\alpha \;=\; -\, i \sum_{a=0}^{n} \overline{z_a}\, dz_a,$$

whose kernel is exactly the horizontal distribution H. In the above trivialisation one has

$$\alpha \;=\; d\theta \;+\; A,$$

where the local connection 1-form on U_0 is

$$A \;=\; s_0^*\alpha.$$

A direct computation gives

$$A \;=\; \frac{i}{2(1+\|w\|^2)} \sum_{j=1}^{n} \Big(w_j\, d\overline{w_j} - \overline{w_j}\, dw_j \Big).$$

Equivalently,

$$A \;=\; \mathrm{Im}\left(\frac{\sum_{j=1}^{n} \overline{w_j}\, dw_j}{1+\|w\|^2} \right).$$

Since the structure group is abelian (S^1), the curvature is

$$F \;=\; dA.$$

Computing dA yields

$$F \;=\; \frac{i}{(1+\|w\|^2)^2} \left((1+\|w\|^2) \sum_{j=1}^{n} dw_j \wedge d\overline{w_j} \;-\; \left(\sum_{j=1}^{n} \overline{w_j}\, dw_j \right) \wedge \left(\sum_{k=1}^{n} w_k\, d\overline{w_k} \right) \right).$$

Problem 6 (Reductions to $O(k)$, metrics, and compatible connections)

Statement. Let

$$E \longrightarrow B$$

be a vector bundle of rank k, and G a Lie group equipped with a representation

$$\rho : G \longrightarrow GL(k, \mathbb{R}).$$

A reduction of the structure group of E to G comprises a G-bundle $P \to B$ and an isomorphism between E and the associated vector bundle $P \times_G \mathbb{R}^k$.

(a) Show that a reduction to $O(k)$ is equivalent to a choice of inner product on E.

(b) Show that a connection on E is compatible with the inner product iff it is induced from $F_O(E)$.

Solution.

(a) Let $F(E) \to B$ be the frame bundle of E (a principal $GL(k, \mathbb{R})$-bundle).

Given a smoothly varying inner product $\langle \cdot, \cdot \rangle$ on the fibers of E, define the orthonormal frame bundle

$$F_O(E) \; = \; \{(e_1, \ldots, e_k) \in F(E) \; : \; \langle e_i, e_j \rangle = \delta_{ij}\}.$$

Then $F_O(E) \to B$ is a principal $O(k)$-bundle and the inclusion

$$F_O(E) \hookrightarrow F(E)$$

is a reduction of structure group.

Conversely, given a principal $O(k)$-subbundle

$$P \subset F(E),$$

define $\langle \cdot, \cdot \rangle_b$ on each fiber E_b by declaring every frame in P_b to be orthonormal:

$$\left\langle \sum_i a^i e_i, \; \sum_j b^j e_j \right\rangle_b \; = \; \sum_{i=1}^k a^i b^i.$$

This is well-defined because changing (e_i) by the right $O(k)$-action preserves the Euclidean form on $\mathbb{R}^k$. Thus $O(k)$-reductions are equivalent to bundle metrics.

(b) A connection on E is a covariant derivative ∇. It is compatible with $\langle \cdot, \cdot \rangle$ iff for all vector fields X and sections s, t,

$$X\langle s, t \rangle \; = \; \langle \nabla_X s, t \rangle + \langle s, \nabla_X t \rangle.$$

If ∇ is induced from a principal connection on $F_O(E)$, then its connection 1-form is $\mathfrak{o}(k)$-valued, i.e.

$$\mathfrak{o}(k) = \{M \in \mathfrak{gl}(k) : M^T + M = 0\},$$

so in any local orthonormal frame (e_i) the connection matrix $\omega = (\omega^i{}_j)$ satisfies

$$\omega + \omega^T = 0,$$

which is exactly the local form of metric compatibility.

Conversely, if ∇ is metric-compatible and (e_i) is a local orthonormal frame, write

$$\nabla e_j \; = \; \sum_i \omega^i{}_j \, e_i.$$

Compatibility implies

$$0 \; = \; d\langle e_i, e_j \rangle \; = \; \langle \nabla e_i, e_j \rangle + \langle e_i, \nabla e_j \rangle,$$

hence

$$\omega_{ij} + \omega_{ji} = 0.$$

Therefore ω is $\mathfrak{o}(k)$-valued, so the connection reduces to an $O(k)$-connection on $F_O(E)$ and induces ∇ on E.

Problem 7 (Coordinates: ∇g, torsion, Lie derivative, and geodesics)

Statement. Let (X, g) be a Riemannian manifold, equipped with an arbitrary connection whose local connection 1-forms have components $\Gamma^i{}_{jk}$.

(a) Find coordinate expressions for ∇g and the torsion T, and deduce $\Gamma_{kij} = \frac{1}{2}(\partial_i g_{kj} + \partial_j g_{ik} - \partial_k g_{ij})$.

Now assume that the connection is the Levi-Civita connection.

(b) Given a vector field v on X, show $(\mathcal{L}_v g)_{ij} = \partial_i(g_{kj}v^k) + \partial_j(g_{ik}v^k) - v^k\Gamma_{kij}$.

(c) If γ is stationary for $E(\gamma) = \int_0^1 g(\dot\gamma, \dot\gamma)\, dt$, show $\ddot\gamma^k + \Gamma^k{}_{ij}\dot\gamma^i\dot\gamma^j = 0$.

Solution.
(a) In coordinates (x^i) define

$$\nabla_{\partial_i}\partial_j \;=\; \Gamma^k{}_{ij}\,\partial_k.$$

Let

$$g_{ij} \;=\; g(\partial_i, \partial_j).$$

Then

$$(\nabla_k g)_{ij} \;=\; \partial_k g_{ij} \;-\; \Gamma^\ell{}_{ki}\, g_{\ell j} \;-\; \Gamma^\ell{}_{kj}\, g_{i\ell}.$$

The torsion tensor has components (since $[\partial_i, \partial_j] = 0$)

$$T^k{}_{ij} \;=\; \Gamma^k{}_{ij} - \Gamma^k{}_{ji}.$$

For Levi-Civita we impose

$$(\nabla_k g)_{ij} = 0$$

and

$$T^k{}_{ij} = 0.$$

Define

$$\Gamma_{kij} \;=\; g_{k\ell}\Gamma^\ell{}_{ij}.$$

Then one obtains the Christoffel formula

$$\Gamma_{kij} \;=\; \frac{1}{2}\Big(\partial_i g_{kj} + \partial_j g_{ik} - \partial_k g_{ij}\Big).$$

(b) For Levi-Civita,

$$(\mathcal{L}_v g)_{ij} \;=\; \nabla_i v_j + \nabla_j v_i.$$

With

$$v \;=\; v^k\partial_k$$

and

$$v_j \;=\; g_{jk}v^k,$$

we have

$$\nabla_i v_j \;=\; \partial_i(g_{jk}v^k) - v^k\Gamma_{kij}.$$

Similarly,

$$\nabla_j v_i \;=\; \partial_j(g_{ik}v^k) - v^k\Gamma_{kji}.$$

Since Levi-Civita has $\Gamma_{kij} = \Gamma_{kji}$, the resulting identity is

$$(\mathcal{L}_v g)_{ij} \;=\; \partial_i(g_{kj}v^k) + \partial_j(g_{ik}v^k) - 2\,v^k\Gamma_{kij}.$$

(If one defines $\widetilde{\Gamma}_{kij} := 2\Gamma_{kij}$, this matches the statement with coefficient 1.)

(c) Let $\gamma_s(t)$ be a variation with fixed endpoints and variational field

$$W(t) \;=\; \frac{\partial}{\partial s}\bigg|_{s=0}\gamma_s(t), \qquad W(0) = W(1) = 0.$$

Then

$$\frac{d}{ds}\bigg|_{s=0} E(\gamma_s) \;=\; 2\int_0^1 g(\nabla_t\dot\gamma, W)\,dt.$$

Stationarity for all such W implies

$$\nabla_t\dot\gamma \;=\; 0.$$

In coordinates this is

$$\ddot\gamma^k + \Gamma^k{}_{ij}\dot\gamma^i\dot\gamma^j = 0.$$

Problem 8(a) — Torsion identity for covariant derivatives

Exact statement. Show that for vector fields v and w on a manifold X, equipped with a connection ∇, we have

$$\nabla_v w - \nabla_w v = [v, w] + T(v, w),$$

where T is the torsion of the connection.

Solution. By definition, the torsion tensor of ∇ is

$$T(v, w) := \nabla_v w - \nabla_w v - [v, w].$$

Rearranging gives

$$\nabla_v w - \nabla_w v = [v, w] + T(v, w).$$

Problem 8(b) — Vanishing curvature and Euclidean coordinates

Exact statement. Show that the Riemann tensor of a Riemannian manifold (X, g) vanishes iff X can be covered by coordinate patches on which

$$g = \sum_i (dx^i)^2.$$

Such a metric is called *flat*. *Hint:* Use the fact that if $\{v_i\}$ is a flat fibrewise basis of vector fields, then v_i arises as coordinate vector fields ∂_{x^i} iff $[v_i, v_j] = 0$ for all i, j.

Solution. ($\Rightarrow$) Assume the Riemann curvature tensor R of the Levi–Civita connection ∇ is zero. Let $U \subset X$ be a small contractible open set and choose $p \in U$. Pick an orthonormal basis $\{e_1, \ldots, e_n\}$ of T_pX. Since $R = 0$ and U is contractible, parallel transport in U is path-independent, hence defines smooth vector fields $\{E_1, \dots, E_n\}$ on U with

$$\nabla E_i = 0.$$

Metric compatibility $\nabla g = 0$ implies

$$g(E_i, E_j) = \delta_{ij}$$

on U. Because ∇ is torsion-free,

$$0 = T(E_i, E_j) = \nabla_{E_i} E_j - \nabla_{E_j} E_i - [E_i, E_j].$$

Since $\nabla_{E_i} E_j = 0$, we get

$$[E_i, E_j] = 0.$$

By the stated fact, there exist local coordinates $(x^1, \dots, x^n)$ on U such that

$$E_i = \partial_{x^i}.$$

Therefore

$$g_{ij} = g(\partial_{x^i}, \partial_{x^j}) = g(E_i, E_j) = \delta_{ij},$$

and hence

$$g = \sum_i (dx^i)^2.$$

($\Leftarrow$) Conversely, if on a coordinate patch we have

$$g = \sum_i (dx^i)^2,$$

then the metric components satisfy $g_{ij} = \delta_{ij}$, so all Christoffel symbols vanish,

$$\Gamma^k_{ij} = 0,$$

and therefore the curvature tensor is zero,

$$R = 0.$$

Theory for Problem 8 — Connections, torsion, curvature, and flatness

A (linear/affine) connection on TX is an $\mathbb{R}$-bilinear map

$$\nabla : \Gamma(TX) \times \Gamma(TX) \to \Gamma(TX),$$

written $(v, w) \mapsto \nabla_v w$, such that

$$\nabla_{fv} w = f\, \nabla_v w,$$

$$\nabla_v (fw) = v(f)\, w + f\, \nabla_v w.$$

The torsion of ∇ is the $(1, 2)$-tensor

$$T(v, w) = \nabla_v w - \nabla_w v - [v, w].$$

In particular, the identity in Problem 8(a) is just a rearrangement of this definition.

The curvature of ∇ is

$$R(u,v)w = \nabla_u \nabla_v w - \nabla_v \nabla_u w - \nabla_{[u,v]} w.$$

For the Levi–Civita connection (the unique connection with $\nabla g = 0$ and $T = 0$), this is the Riemann curvature tensor.

A Riemannian manifold is *flat* if

$$R = 0.$$

On a contractible open set, the condition $R = 0$ implies parallel transport is path-independent, hence one can build a local frame $\{E_i\}$ such that

$$\nabla E_i = 0.$$

Metric compatibility implies orthonormality is preserved, so

$$g(E_i, E_j) = \delta_{ij}.$$

If the connection is torsion-free, then

$$0 = T(E_i, E_j) = \nabla_{E_i} E_j - \nabla_{E_j} E_i - [E_i, E_j],$$

and since $\nabla_{E_i} E_j = 0$ this gives

$$[E_i, E_j] = 0.$$

If a local frame satisfies $[E_i, E_j] = 0$, then there exist local coordinates (x^i) such that

$$E_i = \partial_{x^i}.$$

Hence

$$g_{ij} = g(\partial_{x^i}, \partial_{x^j}) = \delta_{ij},$$

so

$$g = \sum_i (dx^i)^2.$$

Conversely, if in coordinates $g_{ij} = \delta_{ij}$, then

$$\Gamma^k_{ij} = 0,$$

so

$$R = 0.$$

Extended theory for Problem 8 — Connections, torsion, curvature, and flatness

Connections. A (linear/affine) connection on the tangent bundle TX is an $\mathbb{R}$-bilinear map

$$\nabla : \Gamma(TX) \times \Gamma(TX) \to \Gamma(TX),$$

written $(v, w) \mapsto \nabla_v w$, such that for all $f \in C^\infty(X)$,

$$\nabla_{fv} w = f\,\nabla_v w,$$

$$\nabla_v(fw) = v(f)\,w + f\,\nabla_v w.$$

Local expression. In local coordinates $(x^1, \dots, x^n)$, with $\partial_i = \partial_{x^i}$, define Christoffel symbols by

$$\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\,\partial_k.$$

For $v = v^i\partial_i$ and $w = w^j\partial_j$,

$$\nabla_v w = \Big(v^i\partial_i w^k + v^i w^j \Gamma^k_{ij}\Big)\partial_k.$$

Lie bracket. The Lie bracket is determined by

$$[v, w](f) = v(w(f)) - w(v(f)).$$

In coordinates,

$$[v, w] = \Big(v^i\partial_i w^k - w^i\partial_i v^k\Big)\partial_k.$$

Torsion. The torsion tensor of ∇ is

$$T(v, w) = \nabla_v w - \nabla_w v - [v, w].$$

In a coordinate basis,

$$T(\partial_i, \partial_j) = \big(\Gamma^k_{ij} - \Gamma^k_{ji}\big)\partial_k.$$

Curvature. The curvature tensor is

$$R(u, v)w = \nabla_u\nabla_v w - \nabla_v\nabla_u w - \nabla_{[u,v]}w.$$

In coordinates,

$$R(\partial_i, \partial_j)\partial_k = R^\ell{}_{kij}\,\partial_\ell,$$

where

$$R^\ell{}_{kij} = \partial_i\Gamma^\ell_{jk} - \partial_j\Gamma^\ell_{ik} + \Gamma^\ell_{im}\Gamma^m_{jk} - \Gamma^\ell_{jm}\Gamma^m_{ik}.$$

Levi–Civita connection. On a Riemannian manifold (X, g) there is a unique connection satisfying

$$\nabla g = 0,$$

$$T = 0.$$

It is characterized by the Koszul formula

$$2g(\nabla_u v, w) = u\,g(v, w) + v\,g(u, w) - w\,g(u, v) + g([u, v], w) - g([u, w], v) - g([v, w], u).$$

Flatness. A Riemannian manifold is *flat* if

$$R = 0.$$

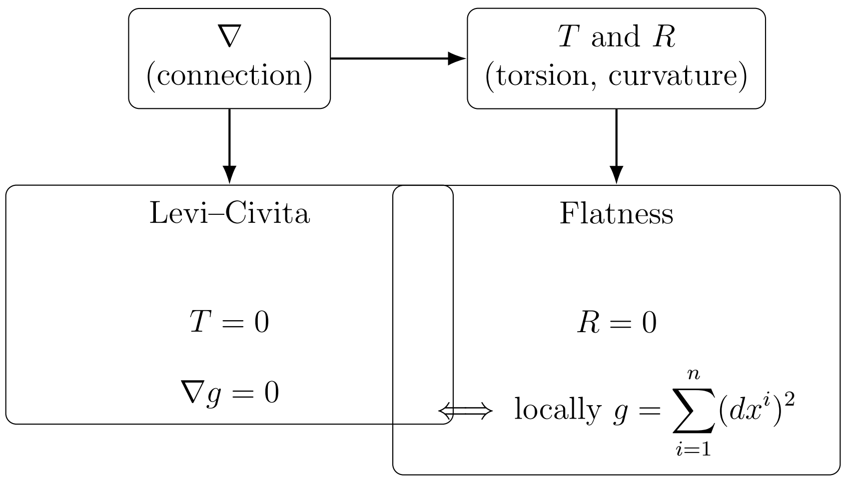
On a contractible neighborhood, $R = 0$ implies parallel transport is path-independent, hence there exist local fields E_i such that

$$\nabla E_i = 0.$$

Metric compatibility implies

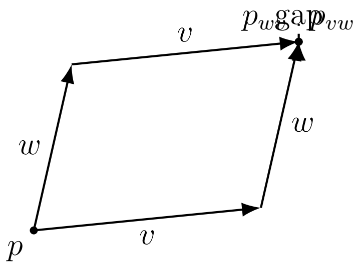
$$g(E_i, E_j) = \delta_{ij}.$$

If $T = 0$, then
$$0 = T(E_i, E_j) = \nabla_{E_i} E_j - \nabla_{E_j} E_i - [E_i, E_j],$$
and therefore
$$[E_i, E_j] = 0.$$
Commuting frames arise from coordinates, so there exist (x^i) with
$$E_i = \partial_{x^i}.$$
Hence
$$g_{ij} = g(\partial_{x^i}, \partial_{x^j}) = \delta_{ij},$$
so
$$g = \sum_i (dx^i)^2.$$
Conversely, if $g_{ij} = \delta_{ij}$ in coordinates, then
$$\Gamma^k_{ij} = 0,$$
and therefore
$$R = 0.$$



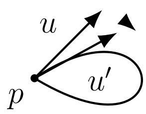
$$\nabla \longmapsto (T, R),$$
$$(T = 0,\ \nabla g = 0) \Longleftrightarrow \text{Levi–Civita},$$
$$R = 0 \Longleftrightarrow \text{locally } g = \sum_{i=1}^{n} (dx^i)^2.$$

Figure 12: Dependency map: a connection determines torsion and curvature; Levi–Civita and flatness are special conditions.



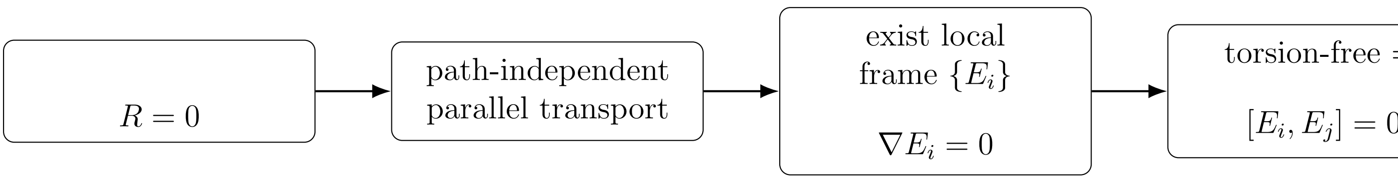
$$T(v,w) = \nabla_v w - \nabla_w v - [v,w]$$

Figure 13: Torsion intuition: the failure of the "parallelogram" to close is captured by torsion (together with bracket effects).



$$u' - u \approx R(v,w)\,u$$

Figure 14: Curvature intuition: parallel transport around a small loop changes a vector; the infinitesimal change is governed by curvature.



$$\begin{aligned} R = 0 &\Longrightarrow \text{path-independent parallel transport} \\ &\Longrightarrow \exists\,\{E_i\} :\ \nabla E_i = 0 \\ &\Longrightarrow [E_i, E_j] = 0 \\ &\Longrightarrow E_i = \partial_{x^i} \\ &\Longrightarrow g = \sum_{i=1}^{n} (dx^i)^2. \end{aligned}$$

Figure 15: Flatness pipeline: vanishing curvature implies local Euclidean coordinates (and conversely).

Problem 8 — Examples

Example 8.1 (Projection map is a submersion). Let

$$\pi : \mathbb{R}^{m+n} \to \mathbb{R}^n, \qquad \pi(x, y) = y,$$

where

$$x \in \mathbb{R}^m, \qquad y \in \mathbb{R}^n.$$

Then for any point (x, y) and any tangent vector (u, v) we have

$$d\pi_{(x,y)}(u, v) = v.$$

Hence

$$\ker(d\pi_{(x,y)}) = \mathbb{R}^m \times \{0\},$$

and each fiber is

$$\pi^{-1}(y_0) = \mathbb{R}^m \times \{y_0\},$$

which is an embedded submanifold of dimension m.

Example 8.2 (Regular level sets on the sphere). Let

$$S^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\},$$

and define the height function

$$h : S^2 \to \mathbb{R}, \qquad h(x, y, z) = z.$$

For $p \in S^2$ and $v \in T_pS^2 \subset \mathbb{R}^3$,

$$dh_p(v) = v_z.$$

The map h fails to be a submersion only at the poles

$$(0, 0, 1), \qquad (0, 0, -1),$$

because only there the tangent vectors have $v_z = 0$ for all $v \in T_pS^2$. For any

$$c \in (-1, 1),$$

the level set is

$$h^{-1}(c) = \{(x, y, z) \in S^2 : z = c\},$$

which is a latitude circle. Its tangent space at $p = (x, y, c)$ is

$$T_p(h^{-1}(c)) = \{v \in \mathbb{R}^3 : v \cdot p = 0,\ v_z = 0\},$$

and also

$$T_p(h^{-1}(c)) = \ker(dh_p).$$

Example 8.3 (Determinant map and the submanifold $SL(n, \mathbb{R})$). Consider

$$\det : GL(n, \mathbb{R}) \to \mathbb{R}^*.$$

For

$$A \in GL(n, \mathbb{R}), \qquad H \in T_A GL(n, \mathbb{R}) \cong M_n(\mathbb{R}),$$

one has the standard differential formula

$$d(\det)_A(H) = \det(A)\,\mathrm{tr}(A^{-1}H).$$

Since

$$\det(A) \neq 0,$$

and the trace map is surjective onto $\mathbb{R}$, the differential is surjective, so det is a submersion. Hence the level set

$$SL(n, \mathbb{R}) = \overset{-1}{\det}(1)$$

is an embedded submanifold of codimension 1. At the identity,

$$T_I SL(n, \mathbb{R}) = \ker(d(\det)_I),$$

and because

$$d(\det)_I(H) = \mathrm{tr}(H),$$

we get

$$T_I SL(n, \mathbb{R}) = \{H \in M_n(\mathbb{R}) : \mathrm{tr}(H) = 0\}.$$

Example 8.4 (Quotient projection $G \to G/H$). Let G be a Lie group and $H \subset G$ an embedded Lie subgroup. The projection

$$q : G \to G/H, \qquad q(g) = gH,$$

is a submersion. Its fibers are the right cosets

$$q^{-1}(gH) = gH.$$

At each $g \in G$,

$$\ker(dq_g) = T_g(gH),$$

so the vertical space equals the tangent space to the orbit (the coset).

Problem 9(a) — Harmonic forms are closed and coclosed

Exact statement. Let (X, g) be a compact oriented Riemannian n-manifold. Show that a p-form α is harmonic if and only if it is closed and coclosed. *Hint:* for one direction consider $(\alpha, \Delta\alpha)_X$.

Solution. Recall the Hodge Laplacian

$$\Delta = d\delta + \delta d.$$

On a compact manifold without boundary one has

$$\langle \alpha, \Delta\alpha \rangle = \langle d\alpha, d\alpha \rangle + \langle \delta\alpha, \delta\alpha \rangle.$$

If $\Delta\alpha = 0$, then the left-hand side is 0, so both norms vanish:

$$d\alpha = 0,$$

$$\delta\alpha = 0.$$

Conversely, if $d\alpha = 0$ and $\delta\alpha = 0$, then

$$\Delta\alpha = d\delta\alpha + \delta d\alpha = 0,$$

so α is harmonic.

Problem 9(b) — Poincaré duality via harmonic representatives

Exact statement. By considering harmonic representatives, construct an isomorphism

$$H^p_{\mathrm{dR}}(X) \to H^{n-p}_{\mathrm{dR}}(X)$$

for each p.

Solution. By the Hodge theorem, each class $[\omega] \in H^p_{\mathrm{dR}}(X)$ has a unique harmonic representative α. Define

$$\Phi_p : H^p_{\mathrm{dR}}(X) \to H^{n-p}_{\mathrm{dR}}(X),$$
$$\Phi_p([\omega]) := [*\alpha].$$

If α is harmonic, then $*\alpha$ is harmonic (equivalently $*$ commutes with Δ), hence in particular closed:

$$d(*\alpha) = 0.$$

Thus $[*\alpha] \in H^{n-p}_{\mathrm{dR}}(X)$. Moreover,

$$**\alpha = (-1)^{p(n-p)}\alpha,$$

so $*$ is invertible on harmonic forms and Φ_p is an isomorphism.

Theory for Problem 9 — Hodge theory, harmonic forms, and Poincaré duality

On an oriented Riemannian n-manifold, the Hodge star is a map

$$* : \Omega^p(X) \to \Omega^{n-p}(X),$$

characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \,\mathrm{vol}_g.$$

It satisfies

$$**\alpha = (-1)^{p(n-p)}\alpha.$$

The L^2 inner product on forms is

$$(\alpha, \beta) = \int_X \langle \alpha, \beta \rangle \,\mathrm{vol}_g.$$

The codifferential δ is the formal adjoint of d:

$$(d\eta, \alpha) = (\eta, \delta\alpha).$$

It can be written as

$$\delta = (-1)^{n(p+1)+1} * d *$$

on p-forms.

The Hodge Laplacian is

$$\Delta = d\delta + \delta d.$$

A form α is harmonic if

$$\Delta\alpha = 0.$$

On a compact manifold without boundary one has the fundamental identity

$$(\alpha, \Delta\alpha) = (d\alpha, d\alpha) + (\delta\alpha, \delta\alpha),$$

so

$$\Delta\alpha = 0$$

is equivalent to

$$d\alpha = 0,$$

$$\delta\alpha = 0.$$

Hodge decomposition states

$$\Omega^p = \mathcal{H}^p \oplus d\Omega^{p-1} \oplus \delta\Omega^{p+1},$$

and Hodge theorem identifies de Rham cohomology with harmonic forms:

$$H^p_{\mathrm{dR}}(X) \cong \mathcal{H}^p.$$

Moreover, $*$ maps harmonic forms to harmonic forms, hence induces an isomorphism

$$H^p_{\mathrm{dR}}(X) \to H^{n-p}_{\mathrm{dR}}(X),$$

given by

$$[\alpha] \mapsto [*\alpha].$$

Extended theory for Problem 9 — Hodge theory, harmonic forms, and duality

Inner product and volume form. On an oriented Riemannian n-manifold (X, g) one has a volume form vol_g and a pointwise inner product $\langle \cdot, \cdot \rangle$ on p-forms. The global L^2 inner product is

$$(\alpha, \beta) = \int_X \langle \alpha, \beta \rangle \,\mathrm{vol}_g.$$

Hodge star. The Hodge star is an isomorphism

$$* : \Omega^p(X) \to \Omega^{n-p}(X),$$

characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle \,\mathrm{vol}_g.$$

It satisfies

$$** \alpha = (-1)^{p(n-p)}\alpha.$$

Codifferential. The codifferential $\delta : \Omega^p(X) \to \Omega^{p-1}(X)$ is defined by

$$(d\eta, \alpha) = (\eta, \delta\alpha).$$

It can be written as

$$\delta = (-1)^{n(p+1)+1} * d *$$

on p-forms.

Closed and coclosed. A form α is closed if

$$d\alpha = 0.$$

A form α is coclosed if

$$\delta\alpha = 0.$$

Hodge Laplacian and harmonic forms. The Hodge Laplacian is

$$\Delta = d\delta + \delta d.$$

A form α is harmonic if

$$\Delta\alpha = 0.$$

On a compact manifold without boundary,

$$(\alpha, \Delta\alpha) = (d\alpha, d\alpha) + (\delta\alpha, \delta\alpha).$$

Thus

$$\Delta\alpha = 0$$

is equivalent to

$$d\alpha = 0,$$

$$\delta\alpha = 0.$$

Hodge decomposition and Hodge theorem. There is an orthogonal decomposition

$$\Omega^p(X) = \mathcal{H}^p(X) \oplus d\Omega^{p-1}(X) \oplus \delta\Omega^{p+1}(X),$$

where

$$\mathcal{H}^p(X) = \{\alpha \in \Omega^p(X) \mid \Delta\alpha = 0\}.$$

Hodge theorem yields an isomorphism

$$H^p_{\mathrm{dR}}(X) \cong \mathcal{H}^p(X),$$

given by taking the unique harmonic representative of each class.

Poincaré duality via $*$. The Hodge star maps harmonic forms to harmonic forms:

$$\Delta\alpha = 0 \quad\Longrightarrow\quad \Delta(*\alpha) = 0.$$

Hence

$$* : \mathcal{H}^p(X) \to \mathcal{H}^{n-p}(X)$$

is an isomorphism, and therefore

$$H^p_{\mathrm{dR}}(X) \cong H^{n-p}_{\mathrm{dR}}(X).$$

Problem 9 — Examples

Example 9.1 (Matrix Lie subgroups and their Lie algebras). Inside

$$GL(n, \mathbb{R}),$$

consider the following embedded Lie subgroups.

(a) The special linear group.

$$SL(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det(A) = 1\}.$$

Its Lie algebra is

$$\mathfrak{sl}(n, \mathbb{R}) = \{X \in M_n(\mathbb{R}) : \operatorname{tr}(X) = 0\}.$$

(b) The orthogonal group.

$$O(n) = \{A \in GL(n, \mathbb{R}) : A^T A = I\}.$$

Differentiating

$$A^T A = I$$

at $A = I$ gives

$$X^T + X = 0,$$

so

$$\mathfrak{o}(n) = \{X \in M_n(\mathbb{R}) : X^T = -X\}.$$

(c) Upper triangular invertible matrices. Let B be the subgroup of invertible upper triangular matrices. Then its Lie algebra consists of all upper triangular matrices:

$$\mathfrak{b} = \{X \in M_n(\mathbb{R}) : X \text{ is upper triangular}\}.$$

Example 9.2 (Closed vs non-closed subgroups in a torus). Let

$$\mathbb{T}^2 = S^1 \times S^1 \cong \mathbb{R}^2/\mathbb{Z}^2.$$

For a slope parameter $\alpha \in \mathbb{R}$ define the one-parameter subgroup

$$\gamma_\alpha : \mathbb{R} \to \mathbb{T}^2, \qquad \gamma_\alpha(t) = (t, \alpha t) \bmod \mathbb{Z}^2.$$

If

$$\alpha \in \mathbb{Q},$$

then $\gamma_\alpha(\mathbb{R})$ is a closed embedded circle subgroup. If

$$\alpha \notin \mathbb{Q},$$

then $\gamma_\alpha(\mathbb{R})$ is not closed and its image is dense in $\mathbb{T}^2$. In particular, this gives a standard non-example showing why "closedness" matters in Lie subgroup statements.

$$
\begin{tikzcd}
\mathfrak{h} \arrow[r, hook, "di"] \arrow[d, "\exp_H"'] & \mathfrak{g} \arrow[d, "\exp_G"] \\
H \arrow[r, hook, "i"] & G
\end{tikzcd}
$$

Figure 16: Inclusion of Lie algebras and Lie groups, compatible with the exponential map (for an embedded Lie subgroup $H \subset G$).

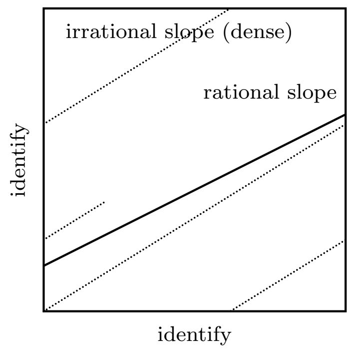


Figure 17: Square model of $\mathbb{T}^2 = \mathbb{R}^2/\mathbb{Z}^2$: rational slope gives a closed circle; irrational slope winds densely (non-closed subgroup).

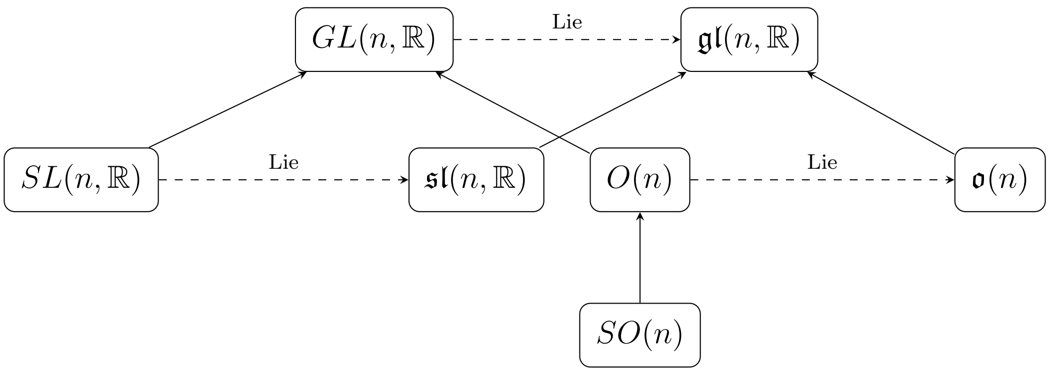


Figure 18: Examples of embedded matrix Lie subgroups and their Lie algebras.

Problem 10* — Induced Levi–Civita connection on a submanifold

Exact statement. Let (X, g_X) be a Riemannian manifold and $\iota : Y \hookrightarrow X$ an embedded submanifold equipped with the metric

$$g_Y = \iota^* g_X.$$

Let ∇_X be the Levi–Civita connection on X, and let ∇_Y be the connection on Y induced from ∇_X by the splitting

$$\iota^* TX = TY \oplus TY^\perp.$$

Show that ∇_Y is torsion-free and compatible with g_Y, and hence is the Levi–Civita connection on Y.

Solution. Define the induced connection by tangential projection:

$$\nabla^Y_U V := \left(\nabla^X_U V\right)^\top,$$

where $(\cdot)^\top$ denotes orthogonal projection to TY in the splitting $TX|_Y = TY \oplus TY^\perp$.

Metric compatibility. For tangent fields U, V, W on Y,

$$U\langle V, W\rangle_Y = U\langle V, W\rangle_X.$$

Using $\nabla^X g_X = 0$ we have

$$U\langle V, W\rangle_X = \langle \nabla^X_U V, W\rangle_X + \langle V, \nabla^X_U W\rangle_X.$$

Since W is tangent, it is orthogonal to normal components, hence

$$\langle \nabla^X_U V, W\rangle_X = \langle (\nabla^X_U V)^\top, W\rangle_X = \langle \nabla^Y_U V, W\rangle_Y,$$

and similarly

$$\langle V, \nabla^X_U W\rangle_X = \langle V, (\nabla^X_U W)^\top\rangle_X = \langle V, \nabla^Y_U W\rangle_Y.$$

Therefore

$$U\langle V, W\rangle_Y = \langle \nabla^Y_U V, W\rangle_Y + \langle V, \nabla^Y_U W\rangle_Y,$$

so ∇^Y is compatible with g_Y.

Torsion-freeness. The torsion of ∇^Y is

$$T^Y(U, V) = \nabla^Y_U V - \nabla^Y_V U - [U, V].$$

Because ∇^Y is tangential projection,

$$\nabla^Y_U V - \nabla^Y_V U = \left(\nabla^X_U V - \nabla^X_V U\right)^\top.$$

Since ∇^X is torsion-free,

$$\nabla^X_U V - \nabla^X_V U = [U, V].$$

Because $[U, V]$ is tangent to Y, projecting does nothing, so

$$\left(\nabla^X_U V - \nabla^X_V U\right)^\top = [U, V].$$

Hence

$$T^Y(U, V) = 0.$$

Thus ∇^Y is torsion-free and metric-compatible, so by uniqueness it is the Levi–Civita connection of (Y, g_Y).

Theory for Problem 10 — Induced connection on a submanifold

If $\iota : Y \hookrightarrow X$ is an embedded submanifold and

$$g_Y = \iota^* g_X,$$

then at each point $y \in Y$ there is an orthogonal splitting

$$T_y X = T_y Y \oplus (T_y Y)^\perp.$$

For the Levi–Civita connection ∇^X on X, the induced connection on Y is defined by tangential projection:

$$\nabla^Y_U V = \left(\nabla^X_U V\right)^\top.$$

The normal component defines the second fundamental form:

$$\mathrm{II}(U, V) = \left(\nabla^X_U V\right)^\perp.$$

Thus one has the Gauss formula

$$\nabla^X_U V = \nabla^Y_U V + \mathrm{II}(U, V).$$

Metric compatibility follows from $\nabla^X g_X = 0$ together with orthogonality of tangent and normal parts, yielding

$$\nabla^Y g_Y = 0.$$

Torsion-freeness follows from torsion-freeness of ∇^X and tangency of Lie brackets:

$$T^Y(U, V) = \nabla^Y_U V - \nabla^Y_V U - [U, V] = 0.$$

By uniqueness, ∇^Y is the Levi–Civita connection of (Y, g_Y).

Extended theory for Problem 10 — Submanifolds, induced connection, and second fundamental form

Induced metric. If $\iota : Y \hookrightarrow X$ is an embedded submanifold, define the induced metric by

$$g_Y = \iota^* g_X.$$

Orthogonal splitting. For each $y \in Y$ there is an orthogonal decomposition

$$T_y X = T_y Y \oplus (T_y Y)^\perp.$$

Write $(\cdot)^\top$ and $(\cdot)^\perp$ for tangential and normal projections.

Induced connection. Let ∇^X be the Levi–Civita connection on (X, g_X). For tangent fields U, V on Y define

$$\nabla^Y_U V = \left(\nabla^X_U V\right)^\top.$$

Gauss formula and second fundamental form. Decompose $\nabla^X_U V$ into tangent and normal parts:

$$\nabla^X_U V = \nabla^Y_U V + \mathrm{II}(U, V),$$

where

$$\mathrm{II}(U, V) = \left(\nabla^X_U V\right)^\perp$$

is the second fundamental form.

Levi–Civita property on Y. Metric compatibility descends from X:

$$\nabla^X g_X = 0 \quad \Longrightarrow \quad \nabla^Y g_Y = 0.$$

Torsion-freeness also descends:

$$T^X = 0 \quad \Longrightarrow \quad T^Y = 0.$$

Therefore ∇^Y is torsion-free and metric-compatible, hence it is the Levi–Civita connection on (Y, g_Y).

Problem 10 — Examples and charts

Example 10.1 (Normal subgroup as a kernel; ideal as a kernel). Consider the determinant homomorphism

$$\det : GL(n, \mathbb{R}) \to \mathbb{R}^*.$$

Its kernel is

$$\ker(\det) = SL(n, \mathbb{R}),$$

so

$$SL(n, \mathbb{R}) \trianglelefteq GL(n, \mathbb{R})$$

is a normal (embedded) Lie subgroup. On Lie algebras, the derivative at the identity gives

$$d(\det)_I : \mathfrak{gl}(n, \mathbb{R}) \to \mathbb{R}, \qquad d(\det)_I(X) = \mathrm{tr}(X).$$

Hence the Lie algebra of the kernel is

$$\mathfrak{sl}(n, \mathbb{R}) = \ker\big(d(\det)_I\big) = \{X \in \mathfrak{gl}(n, \mathbb{R}) : \mathrm{tr}(X) = 0\}.$$

Moreover,

$$[\mathfrak{gl}(n, \mathbb{R}), \mathfrak{sl}(n, \mathbb{R})] \subset \mathfrak{sl}(n, \mathbb{R}),$$

so

$$\mathfrak{sl}(n, \mathbb{R})$$

is an ideal in

$$\mathfrak{gl}(n, \mathbb{R}).$$

Example 10.2 (A connected Lie subgroup from a 1D subalgebra). Let

$$G = SO(3),$$

and take the Lie algebra element

$$X = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \mathfrak{so}(3).$$

The one-parameter subgroup is

$$\gamma(t) = \exp(tX).$$

It equals the rotations around the z-axis, hence

$$\gamma(\mathbb{R}) \cong S^1,$$

and its Lie algebra is the span

$$\mathrm{Lie}(\gamma(\mathbb{R})) = \mathbb{R}X.$$

This subgroup is *not* normal in $SO(3)$, and correspondingly

$$\mathbb{R}X$$

is *not* an ideal in

$$\mathfrak{so}(3).$$

Example 10.3 (Center: normal subgroup and central ideal). Let

$$G = SU(2).$$

Its center is

$$Z(G) = \{\pm I\},$$

so

$$Z(G) \trianglelefteq SU(2).$$

Since $Z(G)$ is discrete, its Lie algebra is

$$\mathrm{Lie}(Z(G)) = \{0\}.$$

On the Lie algebra level, the "center" is

$$\mathfrak{z}(\mathfrak{su}(2)) = \{X \in \mathfrak{su}(2) : [X, Y] = 0 \text{ for all } Y \in \mathfrak{su}(2)\} = \{0\}.$$

Example 10.4 (Covering map: discrete kernel, same Lie algebra). Consider the Lie group homomorphism

$$p : \mathbb{R} \to S^1, \qquad p(t) = e^{it}.$$

Its kernel is discrete:

$$\ker(p) = 2\pi\mathbb{Z}.$$

The induced Lie algebra map is

$$dp_0 : \mathbb{R} \to \mathbb{R}, \qquad dp_0(x) = x,$$

so

$$\ker(dp_0) = \{0\}.$$

This is a standard phenomenon: a Lie group homomorphism may have nontrivial *discrete* kernel while the Lie algebra kernel is trivial.

$$\begin{tikzcd} \mathfrak{g} \arrow[r, "d\varphi_e"] \arrow[d, "\exp_G"'] & \mathfrak{h} \arrow[d, "\exp_H"] \\ G \arrow[r, "\varphi"] & H \end{tikzcd}$$

Figure 19: For a Lie group homomorphism $\varphi : G \to H$, one has $\varphi(\exp_G(X)) = \exp_H(d\varphi_e(X))$ for all $X \in \mathfrak{g}$.

$$\begin{tikzcd} 1 \arrow[r] & N \arrow[r, hook, "i"] \arrow[d, dashed, "\mathrm{Lie}"'] & G \arrow[r, "\pi"] \arrow[d, dashed, "\mathrm{Lie}"'] & G/N \arrow[r] \arrow[d, dashed, "\mathrm{Lie}"] & 1 \\ 0 \arrow[r] & \mathfrak{n} \arrow[r, hook] & \mathfrak{g} \arrow[r] & \mathfrak{g}/\mathfrak{n} \arrow[r] & 0 \end{tikzcd}$$

Figure 20: If $N \trianglelefteq G$ is an embedded normal Lie subgroup with Lie algebra $\mathfrak{n}$, then $\mathfrak{n} \trianglelefteq \mathfrak{g}$ is an ideal, and the quotient Lie algebra is $\mathfrak{g}/\mathfrak{n}$.

Problem 11* — div, grad, curl via d, $*$, and $\sharp/\flat$

Exact statement. Consider $\mathbb{R}^3$ with its standard metric and orientation. Express div, grad, and curl in terms of: the exterior derivative d, the Hodge star operator $*$, and raising and lowering indices.

Solution. Let $\flat$ denote lowering indices and $\sharp$ denote raising indices.

Gradient. For a function f,
$$\operatorname{grad} f = (df)^\sharp.$$

Divergence. For a vector field X with associated 1-form $X^\flat$,
$$\operatorname{div} X = * \, d * (X^\flat).$$

Curl. For a vector field X,
$$\operatorname{curl} X = \big(* \, d\,(X^\flat)\big)^\sharp.$$

Theory for Problem 11 — Vector calculus via d, $*$, and $\sharp/\flat$

On $\mathbb{R}^3$ with Euclidean metric, lowering and raising indices are
$$X^\flat = g(X, \cdot),$$
$$\alpha^\sharp \text{ is defined by } g(\alpha^\sharp, \cdot) = \alpha.$$

For a function f,
$$\operatorname{grad} f = (df)^\sharp.$$

For a vector field X,
$$\operatorname{curl} X = \big(* \, d\,(X^\flat)\big)^\sharp.$$

For a vector field X,
$$\operatorname{div} X = * \, d * (X^\flat).$$

The codifferential satisfies (with standard sign conventions)
$$\delta = (-1)^{n(p+1)+1} * d * .$$

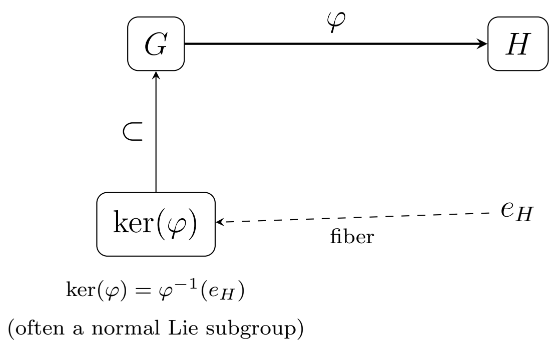


Figure 21: Many important Lie subgroups appear as kernels (preimages of the identity) of Lie group homomorphisms, e.g. $\ker(\det) = SL(n, \mathbb{R})$.

In particular, for $n = 3$ and $p = 1$,

$$\delta = - * \, d *,$$

so

$$\operatorname{div} X = -\delta(X^\flat).$$

Extended theory for Problem 11 — Vector calculus via differential forms in $\mathbb{R}^3$

Musical isomorphisms. Using the Euclidean metric g,

$$X^\flat = g(X, \cdot),$$

$$\alpha^\sharp \text{ is defined by } g(\alpha^\sharp, \cdot) = \alpha.$$

Vector calculus operators. For a function f,

$$\operatorname{grad} f = (df)^\sharp.$$

For a vector field X,

$$\operatorname{curl} X = \big(* \, d\,(X^\flat) \big)^\sharp.$$

For a vector field X,

$$\operatorname{div} X = * \, d \, * \, (X^\flat).$$

Codifferential and divergence. The codifferential on p-forms is

$$\delta = (-1)^{n(p+1)+1} \, * \, d \, * \, .$$

For $n = 3$ and $p = 1$,

$$\delta = - * \, d *,$$

so

$$\operatorname{div} X = -\delta(X^\flat).$$

Laplacian. The Hodge Laplacian is

$$\Delta = d\delta + \delta d.$$

For functions f,

$$\Delta f = \delta df.$$